

Summary of previous class

- If deep seasonal mixing (turnover) stops in a temperate lake due to unseasonably warm fall temperatures, what happens to the oxygen levels at the bottom hypolimnion layer, and how does that impact the ecosystem?

Surface water fails to cool to 4°C (water's maximum density), preventing the lake from achieving a uniform density profile. Without turnover, atmospheric oxygen cannot mix into the deep hypolimnion. Bacteria continue decomposing organic matter at the bottom, depleting the remaining dissolved oxygen and creating a hypoxic or anoxic (dead) zone. This forces deep-water fish to abandon their cold-water habitat and can trigger toxic algal blooms when trapped bottom nutrients (like phosphorus) eventually mix upward.

- Why are unconfined aquifers significantly more vulnerable to surface contamination than confined aquifers, and how does karst topography accelerate this risk?

Unconfined aquifers lack an overlying impermeable rock layer (aquitard), making them exposed to surface runoff, agricultural chemicals, and spills that seep directly through porous soil. Karst topography accelerates this vulnerability because soluble bedrock (like limestone) dissolves into open sinkholes, caves, and underground channels. Contaminants bypass natural soil and sand filtration entirely, flowing rapidly through these underground pipes into the water table without purification.

- Why does the melting of floating sea ice fail to raise sea levels directly, while the melting of land-based ice sheets causes direct sea-level rise?

Floating sea ice is already displacing its own mass in the ocean (Archimedes' Principle); when it melts, the resulting freshwater volume equals the volume of sea ice that was submerged, causing zero net direct sea-level change. Land-based ice sheets (e.g., Greenland, Antarctica) sit on solid rock above sea level. When they melt, water previously stored outside the ocean basin flows into the ocean, adding new mass to the global water volume and driving direct sea-level rise.



Summary of previous class

- When sea ice forms at high latitudes, what happens to the salinity and density of the surrounding surface ocean water through brine rejection, and how does this drive vertical ocean mixing?

As seawater freezes, its crystal lattice rejects salt ions, expelling concentrated salt into the surrounding unfrozen surface water. This process brine rejection dramatically increases the surface salinity and density of the remaining liquid water. Because this water is both cold and hypersaline, it becomes extremely dense and sinks rapidly toward the seafloor, driving vertical ocean mixing and creating deep polar water masses (e.g., North Atlantic Deep Water).

- How does an aquitard differ from an aquifer in terms of hydraulic conductivity, and how does an aquitard protect deep groundwater reserves?

An aquitard consists of low-permeability materials (like clay) with very low hydraulic conductivity, meaning water moves through it extraordinarily slowly compared to a high-permeability aquifer (sand or gravel). An aquitard acts as a physical barrier; when positioned above a deep aquifer, it prevents surface contaminants and polluted groundwater from migrating downward into the deeper freshwater supply.

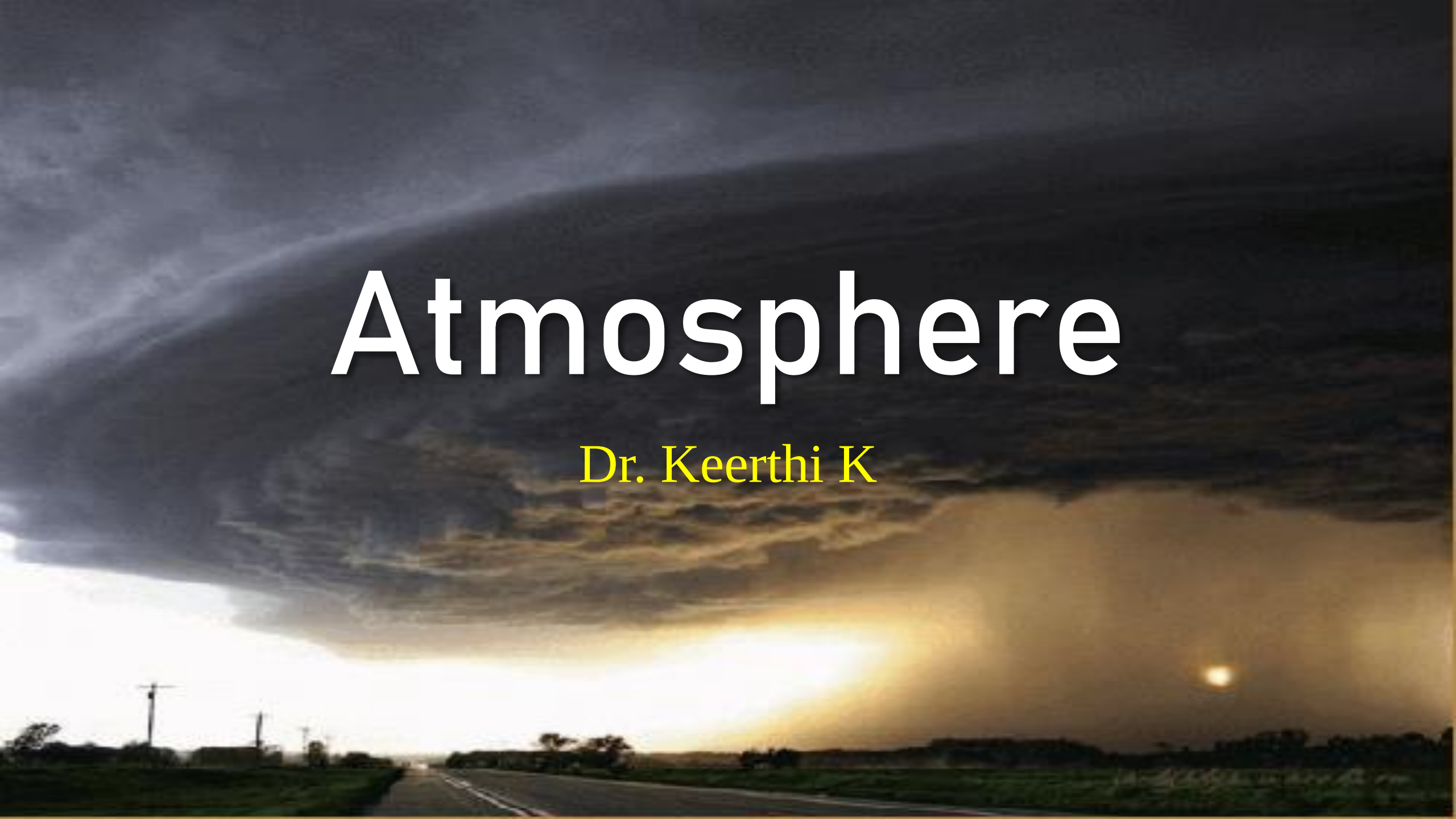
- What is the difference between the active layer and perennially frozen ground in permafrost, and why does thawing permafrost present both structural and environmental hazards?

The active layer is the uppermost layer of soil that thaws during summer and refreezes in winter, whereas perennially frozen ground stays at or below 0 °C continuously for two or more years. Thawing permafrost destabilizes the ground above it, causing soil collapse that damages surface infrastructure (roads, building foundations, pipelines). Environmentally, it releases trapped organic matter that decomposes into greenhouse gases (CO₂ and CH₄), creating a positive climate feedback loop.



Atmosphere

Dr. Keerthi K



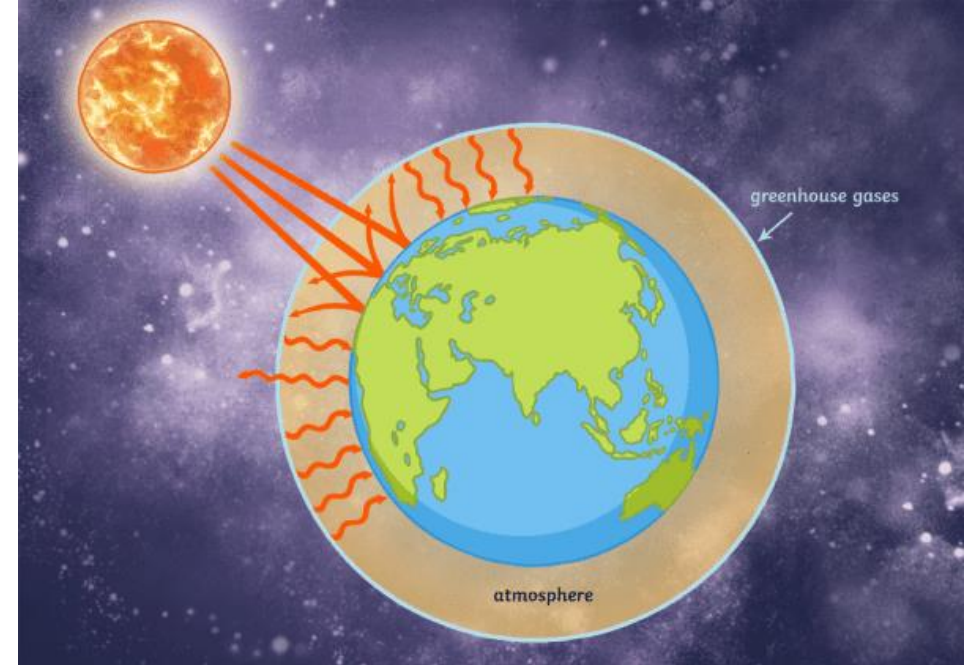
Introduction

- Earth is surrounded by air.
- There is a difference between *atmosphere* and *air*, although we commonly use the words interchangeably.
- An **atmosphere** is a gaseous envelope that surrounds a planet or any other celestial body; Earth is one planet among many that has an atmosphere.
- Air, by contrast, is the **invisible, odorless mixture of gases and suspended particles** that surrounds one special planet— Earth.

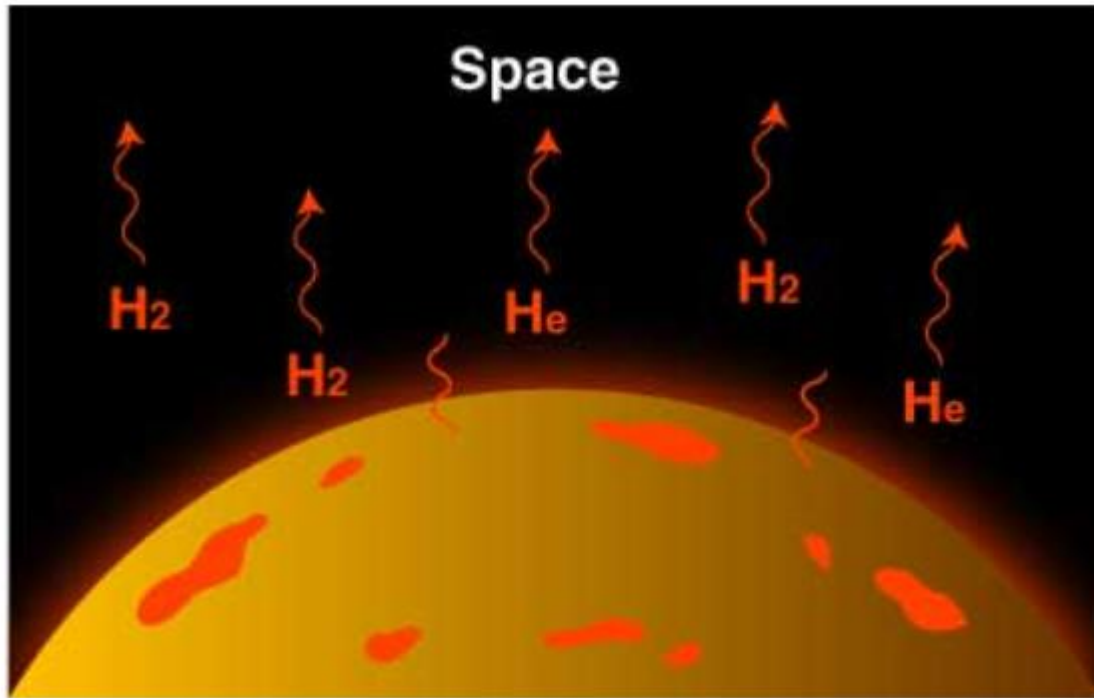


Why is atmospheric composition important ?

- Affects **Earth's surface temperature – greenhouse effect** because of the presence of gases like water vapor and carbon dioxide.
- What we breathe – **oxygen** allows us to exist but atmosphere also contains reactive gases (like ozone) which will react with tissues in your lungs and affect health.
- **Ozone** higher in atmosphere protects life on earth by absorbing UV rays which can damage cells.

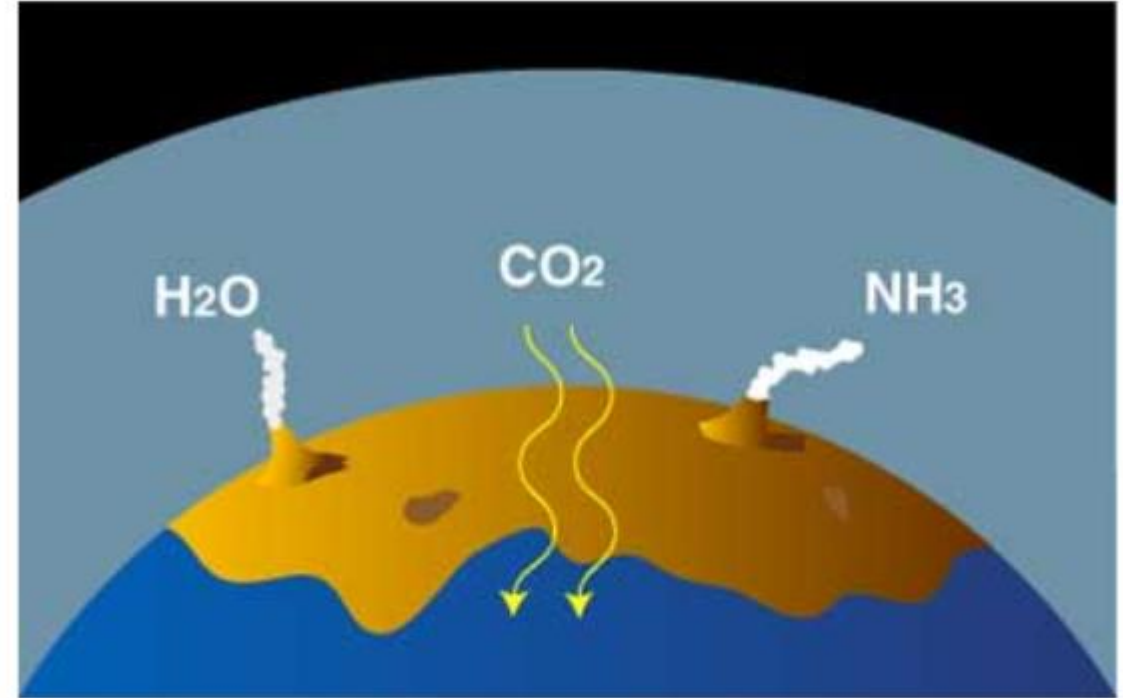


How did Earth's atmosphere form?



Atmosphere #1

Just formed Earth: Like Earth, the hydrogen (H_2) and helium (He) were very warm. These molecules of gas moved so fast they escaped Earth's gravity and eventually all drifted off into space.



Atmosphere #2

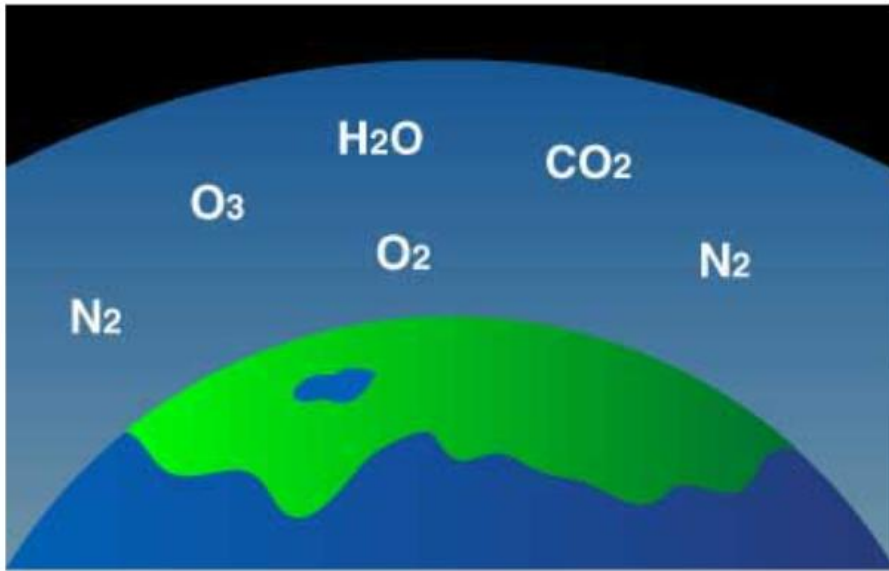
Young Earth: Volcanoes released gases H_2O (water) as steam, carbon dioxide (CO_2), and ammonia (NH_3). Carbon dioxide dissolved in seawater. Simple bacteria thrived on sunlight and CO_2 . By-product is oxygen (O_2).

Composition of volcanic gases

Volcanic Gas Species	Volume, Ave. Subduction Zone (mol%)	Volume, Ave. Rift Zone (mol%)	Mt. Pinatubo (1991), Total Mass (Mt)
H ₂ O	95.4	63.6	250-500 †
CO ₂	6.37	20.6	82-104 ‡
SO ₂	3.72	9.49	15-19
H ₂ S	1.55	0.91	1.2-1.6 ‡
H ₂	1.19	4.91	0.2-0.5 ‡
HCl	0.983	-	0-3
HF	0.097	-	-
CO	0.024	0.92	-
OCS	0.014	0.00071	-

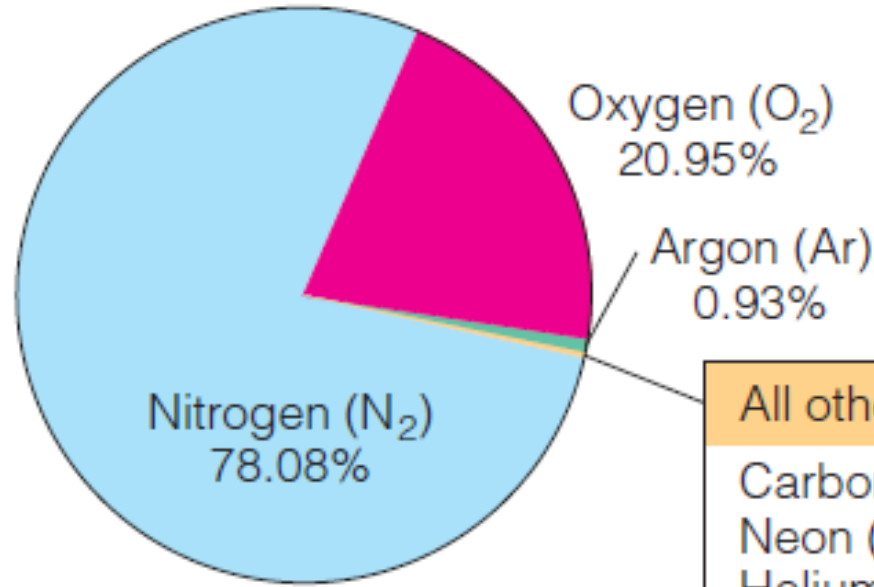


Composition of atmosphere today



Atmosphere #3

Current Earth: Plants and animals thrive in balance. Plants take in carbon dioxide (CO₂) and give off oxygen (O₂). Animals take in oxygen (O₂) and give off CO₂. Burning stuff also gives off CO₂.



Variable substances	
Water vapor (H ₂ O)	1-4%
Aerosols	variable

All other gases, 0.04%	
Carbon dioxide (CO ₂)	0.035%
Neon (Ne)	0.0018%
Helium (He)	0.00052%
Methane (CH ₄)	0.00014%
Krypton (Kr)	0.00010%
Nitrous oxide (N ₂ O)	0.00005%
Hydrogen (H ₂)	0.00005%
Ozone (O ₃)	0.000007%

What happened to atmospheric H₂O ?
What happened to atmospheric CO₂ ?
Where did the O₂ come from ?

- What happened to atmospheric H_2O ?
- The atmosphere can only hold small fraction of the water vapor injected into it from volcanoes. Water vapor condensed into clouds and over time formed the oceans.
- What happened to atmospheric CO_2 ?
- CO_2 is removed from the atmosphere by Biosphere, weathering. Overtime, carbon was locked away in minerals and rocks (and fossil fuels).
- Where did the O_2 come from ?
- Half of the world's oxygen is produced via phytoplankton photosynthesis.
- Solar UV radiation split water vapor in the upper atmosphere into free hydrogen and oxygen. The light hydrogen gas escaped to space, while trace oxygen remained behind.
- The other half is produced via photosynthesis on land by trees, shrubs, grasses, and other plants. As green plants die and fall to the ground or sink to the ocean floor, a small fraction of their organic carbon is buried.



Where did N₂ come from ?

- Released in small amounts by volcanoes but
 - Is chemically inert
 - Is not very soluble in sea water.
- Amount of nitrogen removed from atmosphere was small.
- Over time, with the removal of carbon dioxide – nitrogen became the most abundant gas in the atmosphere.

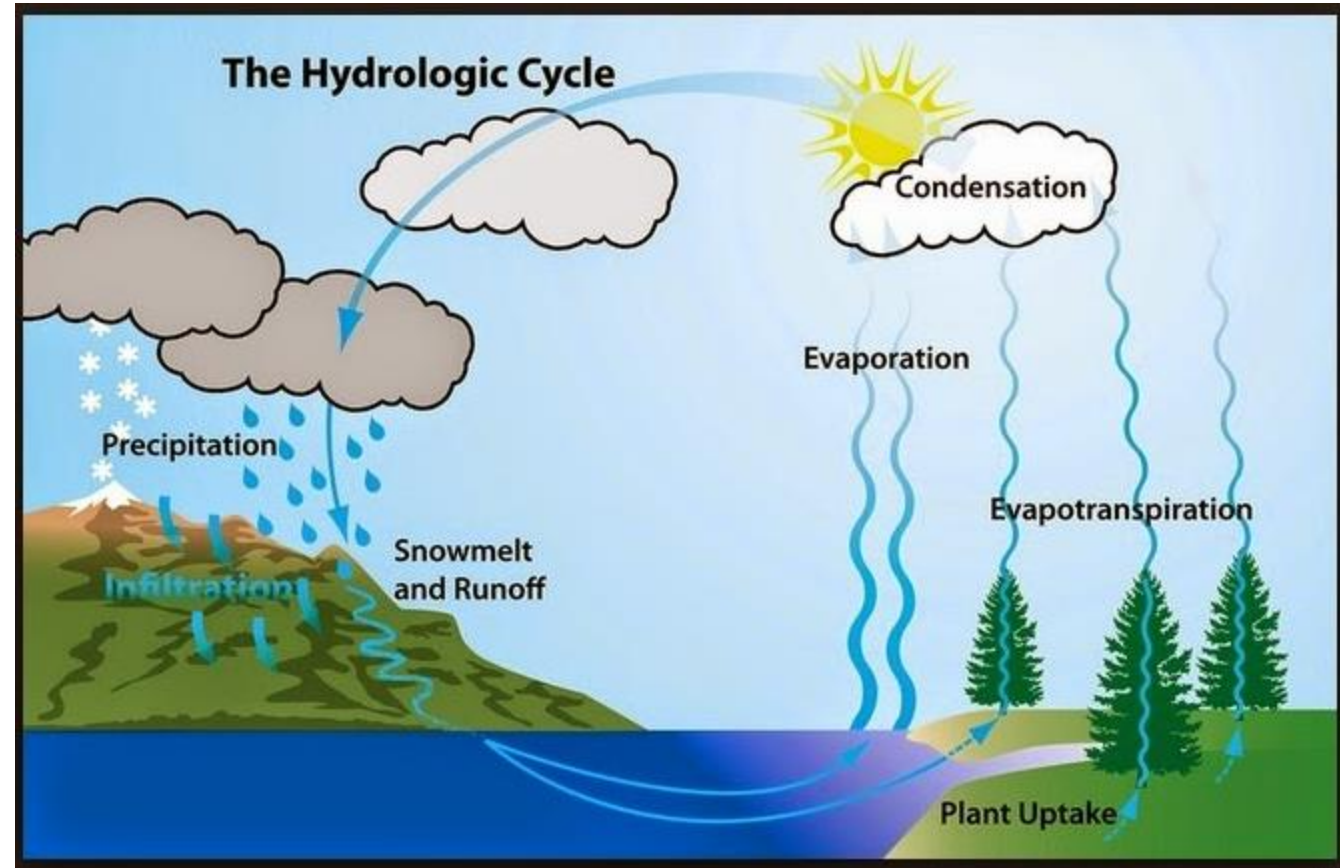
Where did Ar come from ?

- Radioactive decay in the rocks of the planet added Ar to the atmosphere.
- Like nitrogen it is chemically inert and insoluble so has built up through time in the atmosphere.



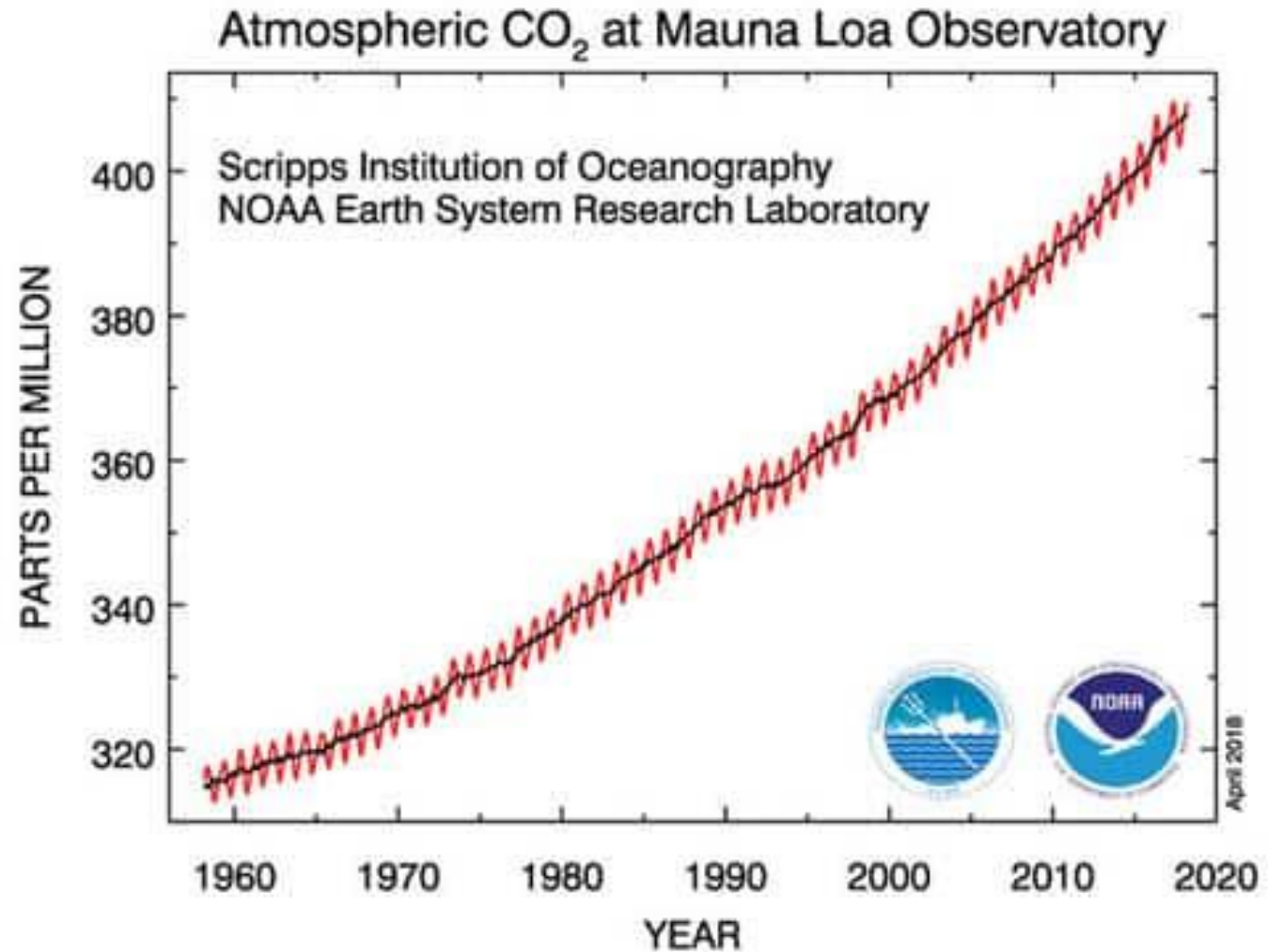
Variable gases: Water vapor

- Most abundant variable gas.
- Supplied to atmosphere by evaporation from the surface.
- Removed from atmosphere by condensation (clouds and rain).
- Concentration is maximum near surface in the tropics and decreases rapidly toward higher altitudes and latitudes.
- A greenhouse gas that absorbs thermal energy emitted by Earth.



Variable gases: CO₂

- Its called as Keeling curve
- Increasing (1.8 ppm/year)
- Wiggles = annual seasons
- Sources = respiration, volcanoes, anthropogenic combustion
- Sinks = photosynthesis, dissolving in sea



Aerosols

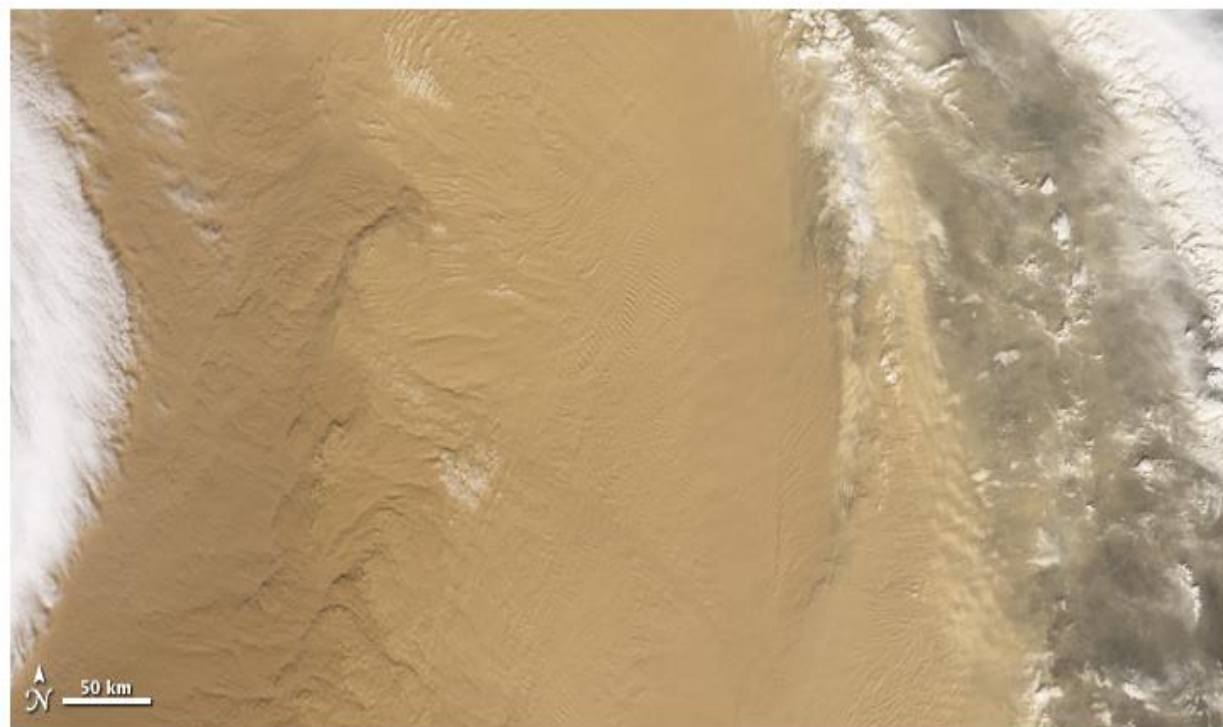
- The air is full of tiny, floating dust particles and droplets - known as “aerosols”. Many aerosols are produced by nature.
- These include sand from desert dust storms, salt particles from the ocean, ice crystals, ash from fires and volcanic eruptions.



Desert dust, volatile organic compounds from vegetation, smoke from forest fires, and volcanic ash

Aerosols

- Depending on the season and weather conditions, aerosols can make their way into the atmosphere almost anywhere on Earth.
- In the Northern Hemisphere, plumes of mineral dust swirl over deserts and arid regions.
- In the Southern Hemisphere, slash-and-burn agriculture in the Amazon and Central Africa releases large amounts of smoke and soot.



Aerosols and Incoming Sunlight (Direct Effects)

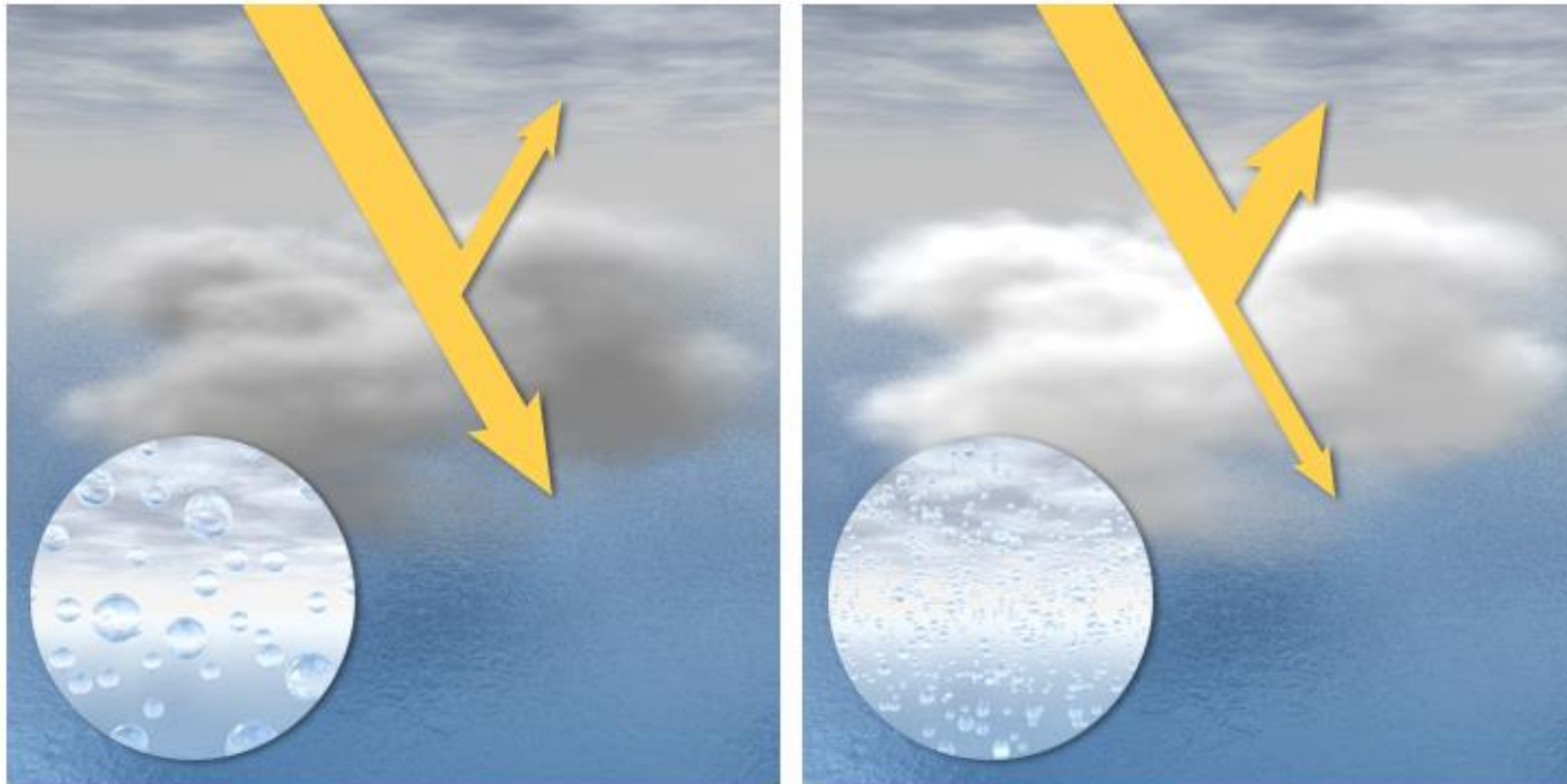
- Different aerosols scatter or absorb sunlight to varying degrees, depending on their physical properties. (Ex: soot – adsorbs, sulfates and nitrates – reflects)
- Climatologists describe these scattering and absorbing properties as the “direct effect” of aerosols on Earth’s radiation field.



Dark aerosols dramatically change the reflectivity of the Earth's surface when they land on snow. Black ash covered the summit of New Zealand's Mount Ruapehu after an eruption in 2007, but was soon covered by fresh snow. Long-term accumulation of black carbon aerosols in the Arctic and Himalaya is leading to increased melting of snow.

Aerosols and Clouds (Indirect Effects)

- Clouds in clean air are composed of a relatively small number of large droplets (left). As a consequence, the clouds are somewhat dark and translucent.
- In air with high concentrations of aerosols, water can easily condense on the particles, creating a large number of small droplets (right). These clouds are dense, very reflective, and bright white.



Aerosols and Clouds

- Water vapor cannot easily condense into liquid droplets on its own in clean air. It requires microscopic solid or liquid particles suspended in the atmosphere, known as **Aerosols** or **Cloud Condensation Nuclei (CCN)**.
- **Providing a Physical Surface:** Pure water vapor molecules move too fast and bounce off one another. Aerosols act as physical landing pads ("seeds") that water molecules can attach to, overcoming the surface tension barrier required to begin droplet growth.
- **Lowering Energy Barriers (Hygroscopic Aerosols):** Many aerosols (like sea salt or sulfates) are *hygroscopic*, meaning they naturally attract water. These particles dissolve slightly as water vapor collects on them, lowering the required humidity level for condensation to begin.
- **Determining Cloud Properties:**
 - **Clean Air (Few Aerosols):** Water vapor condenses onto a small number of seeds, forming fewer, larger water droplets. These droplets fall easily as rain.
 - **Polluted/Aerosol-Rich Air (Many Aerosols):** The same amount of water vapor spreads across millions of tiny seeds, creating a dense cloud made of many smaller droplets. This makes the cloud **brighter and more reflective** (reflecting more solar radiation back into space) and often delays rainfall because the droplets are too small and light to fall.

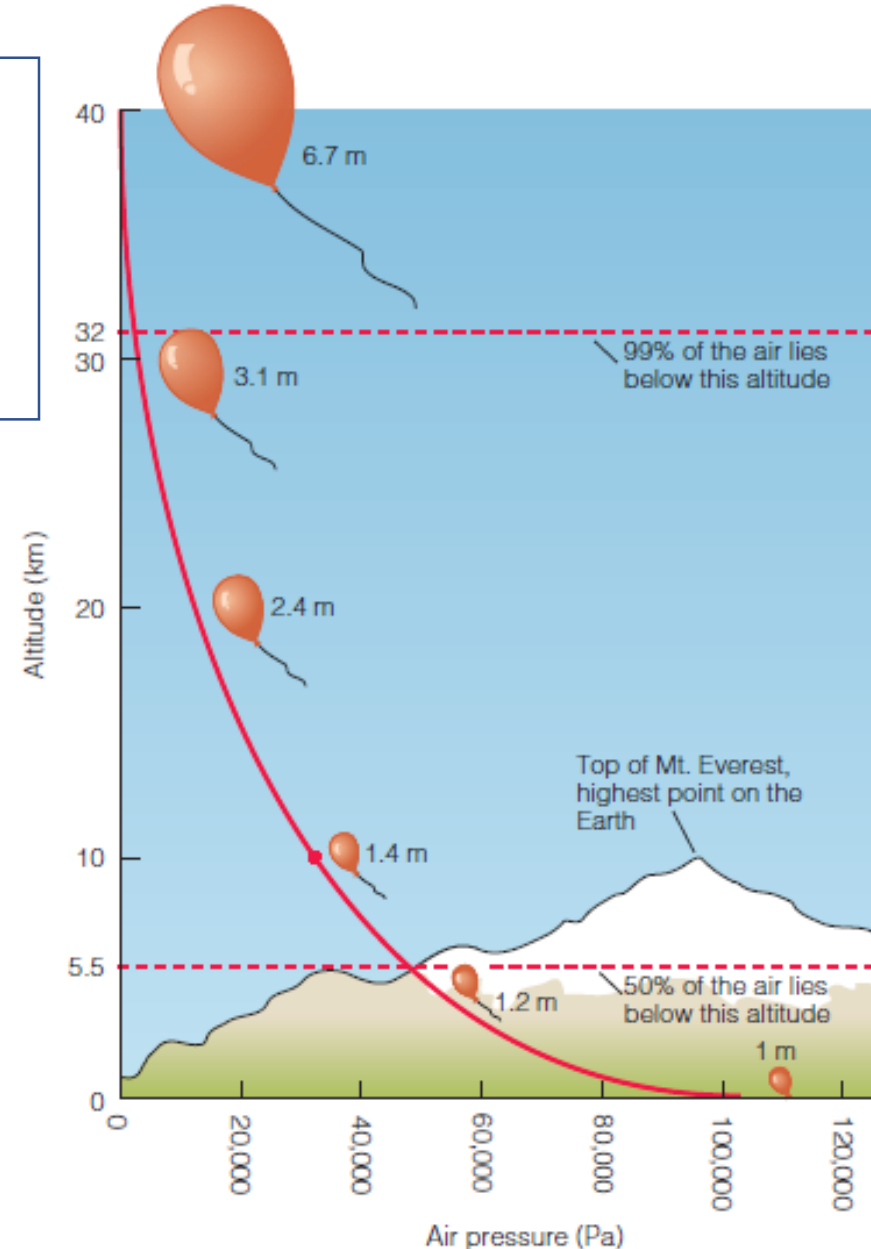
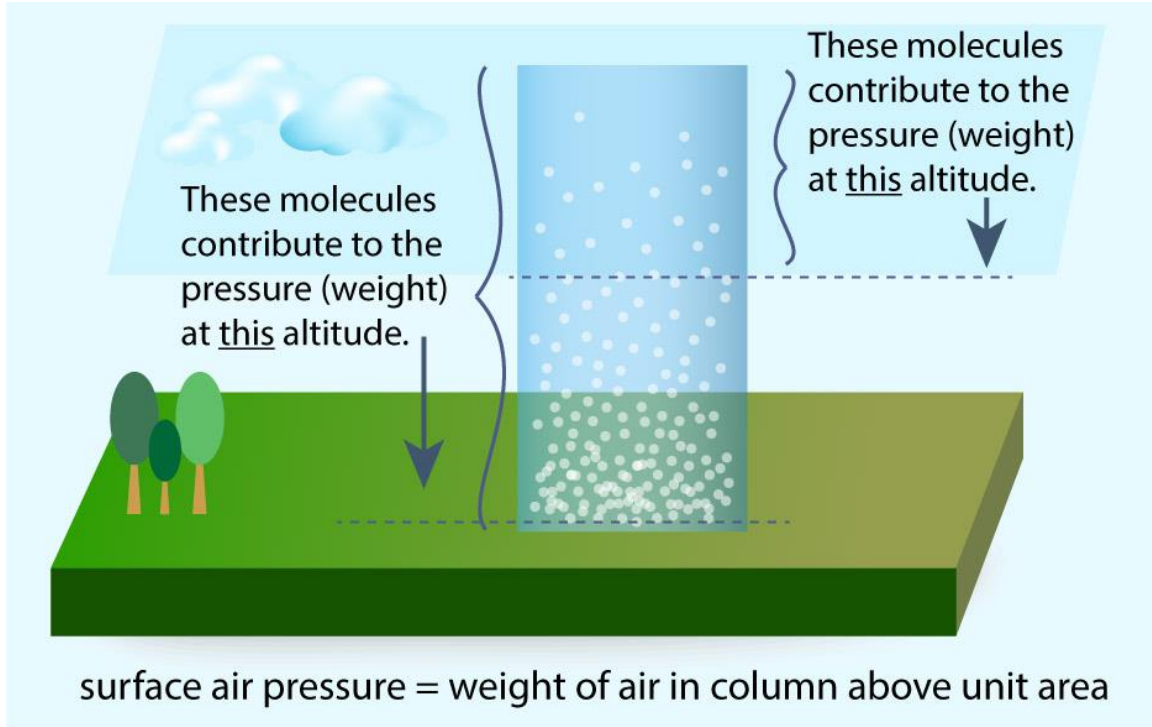


Structure of atmosphere: Pressure

- With increasing altitude
 - Air pressure decreases (compressibility of air)
 - Air density decreases

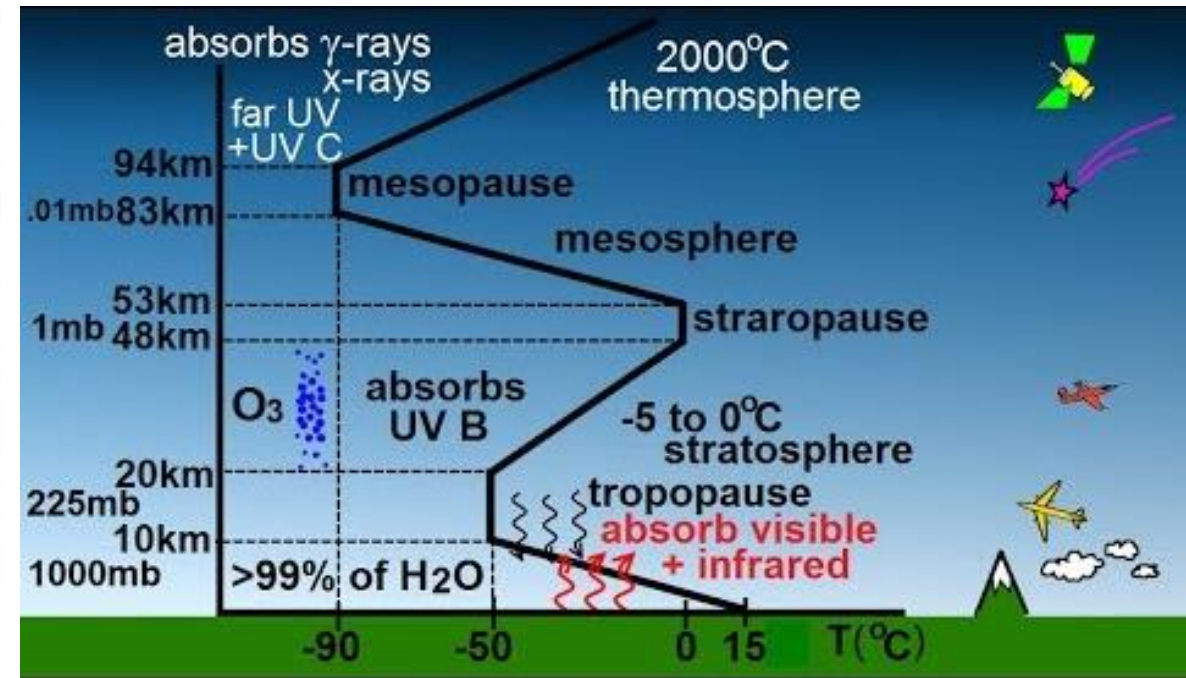
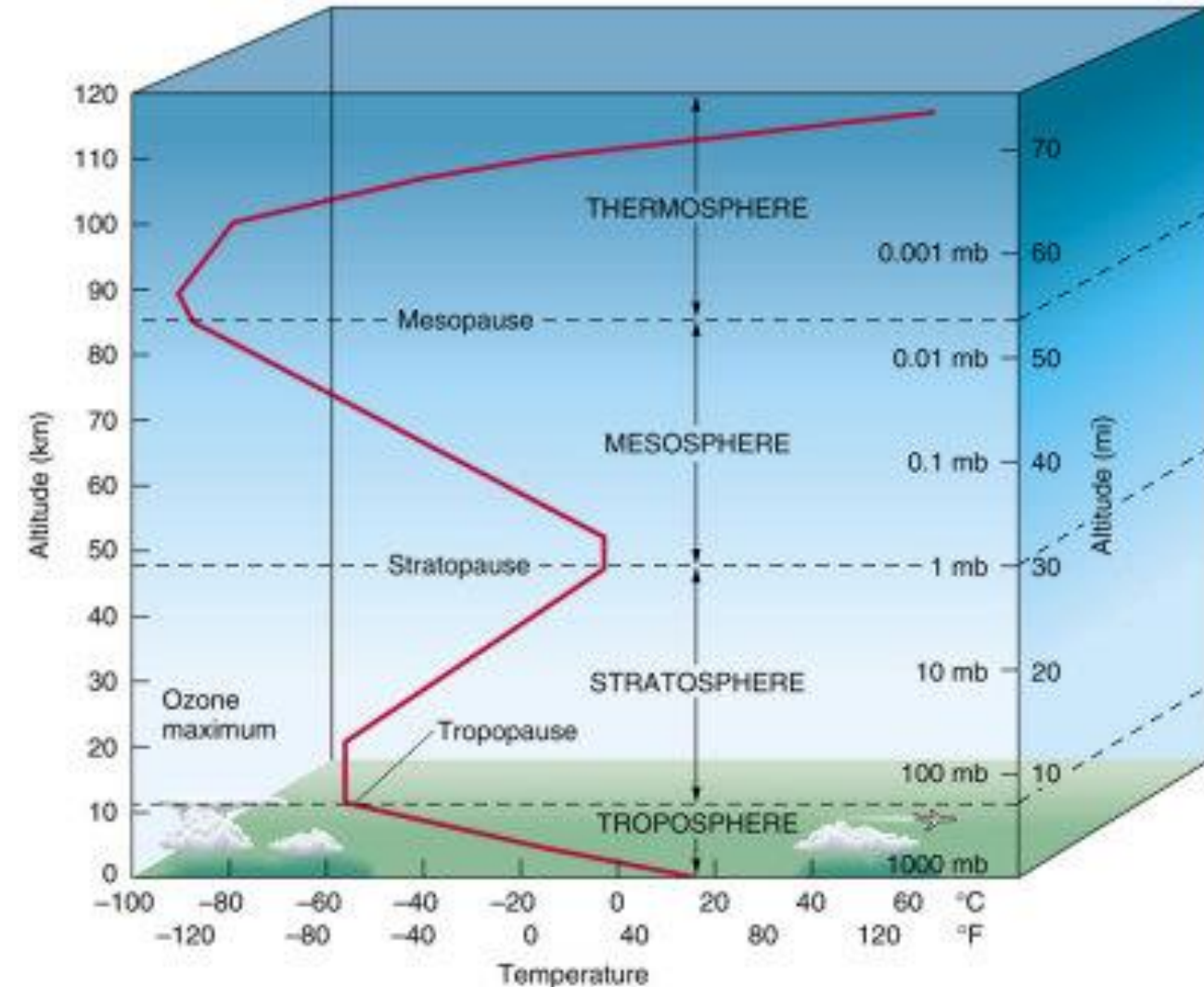
Why ?

- **Gravitational pull**



Structure of atmosphere: Temperature

The atmosphere is **layered**, corresponding with how the atmosphere's temperature changes with altitude.



Why does the air between the troposphere and stratosphere not mix ?

Troposphere

- The temperature of the **troposphere** is highest near the surface of the Earth and decreases with altitude.
- On average, the temperature gradient of the troposphere is **6.5 °C per 1,000 m** (3.6 F per 1,000 ft.) of altitude.

What is the source of heat for the troposphere?

- Earth's surface is a major source of heat for the troposphere, although nearly all of that heat comes from the Sun.
- Sometimes there is a temperature **inversion**, air temperature in the troposphere increases with altitude and warm air sits over cold air. Inversions are very stable and may last for several days or even weeks.

Since temperature inversions are stable, they often trap pollutants and produce unhealthy air conditions in cities. At the top of the troposphere is a thin layer in which the temperature does not change with height. This means that the cooler, denser air of the troposphere is trapped beneath the warmer, less dense air of the stratosphere. Air from the troposphere and stratosphere rarely mix.



Why air moves ?

Atmospheric Convection

- Convection is the circulation of fluid due to density differences.
- Atmospheric convection works like a pot of boiling soup, warm fluid rising (in middle) and cool fluid falling (on sides).
- A rising storm thunderhead is an example of atmospheric convection.
- Warm moist air rises, expands, releases energy as clouds form. After releasing its heat and moisture, the cooled air sinks, displacing warm air below



Winds: moving air

Influenced by

- **Pressure Gradients**

- stronger the pressure gradient the faster the winds

- **Coriolis Force**

- an apparent force, due to Earth's rotation, that changes the direction of moving objects
 - Acts at 90° to the right or left of the direction of travel

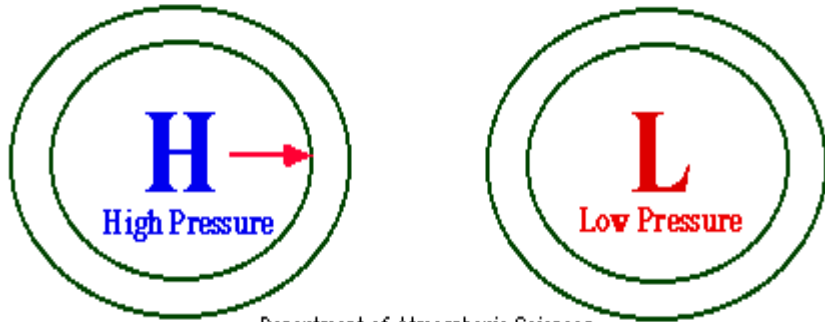
- **Friction**

- resistance to movement; reduces wind speed
 - much greater friction close to Earth surface so wind is slower near surface than in atmosphere
 - the slower the wind the less is the Coriolis effect

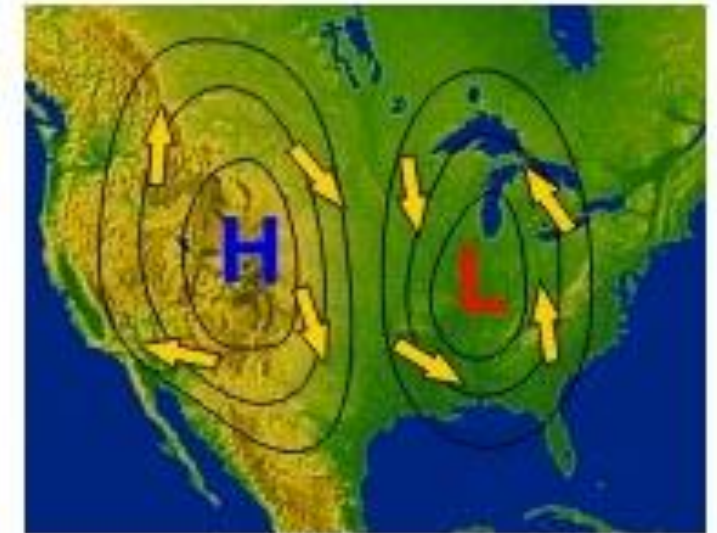
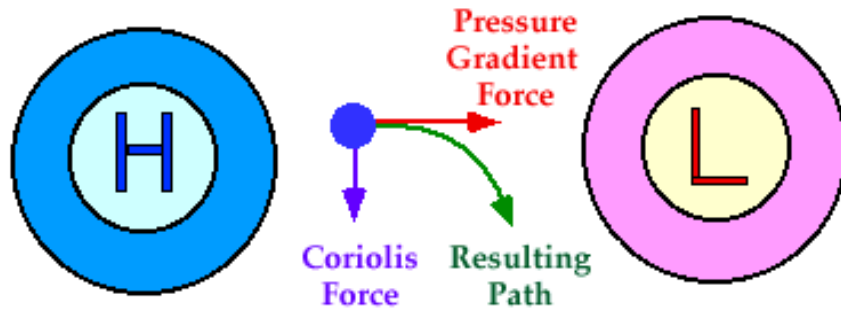


Winds

The influence of the Pressure Gradient Force



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Air moves from areas where high pressure develops to locations where low pressure exit. However, rotation of the Earth results in rotating air masses. Because the Earth's atmosphere is an open system with uneven heating and cooling, masses are constantly forming and changing.

NOAA

High in atmosphere (ignore friction)

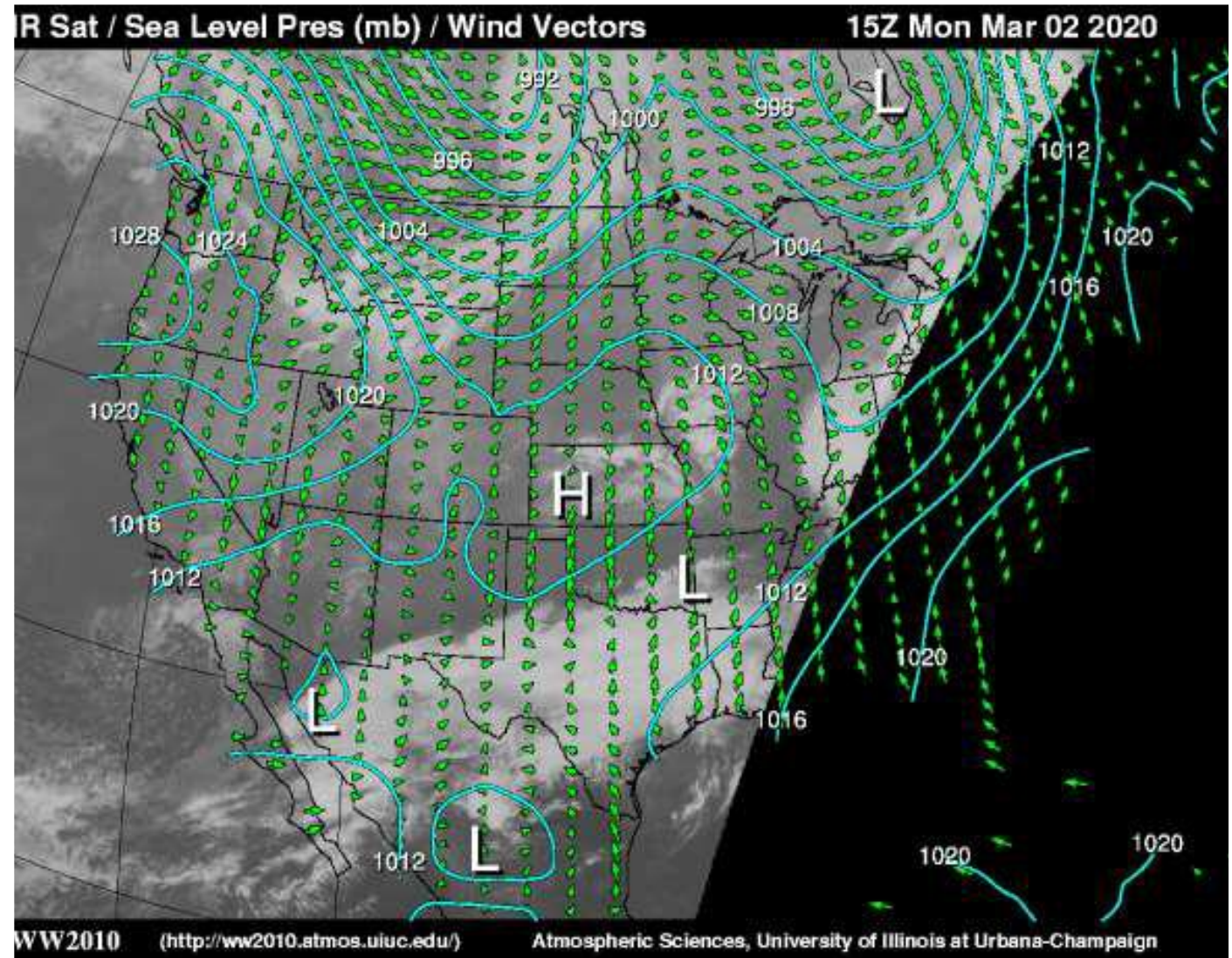
Geostrophic Wind = Pressure Gradient and Coriolis force balance out
= wind blows parallel to isobars



Example Question – 1

The map below depicts sea-level pressure and surface wind vectors. The contours represent pressure contours (isobars) in millibars. The isobars have an interval of 4 millibars. Wind vectors show wind direction and wind speed. The wind speed is directly related to the distance between the isobars. In the background, infrared satellite data shows the cloud patterns over the region. Answer the following questions based on the map.

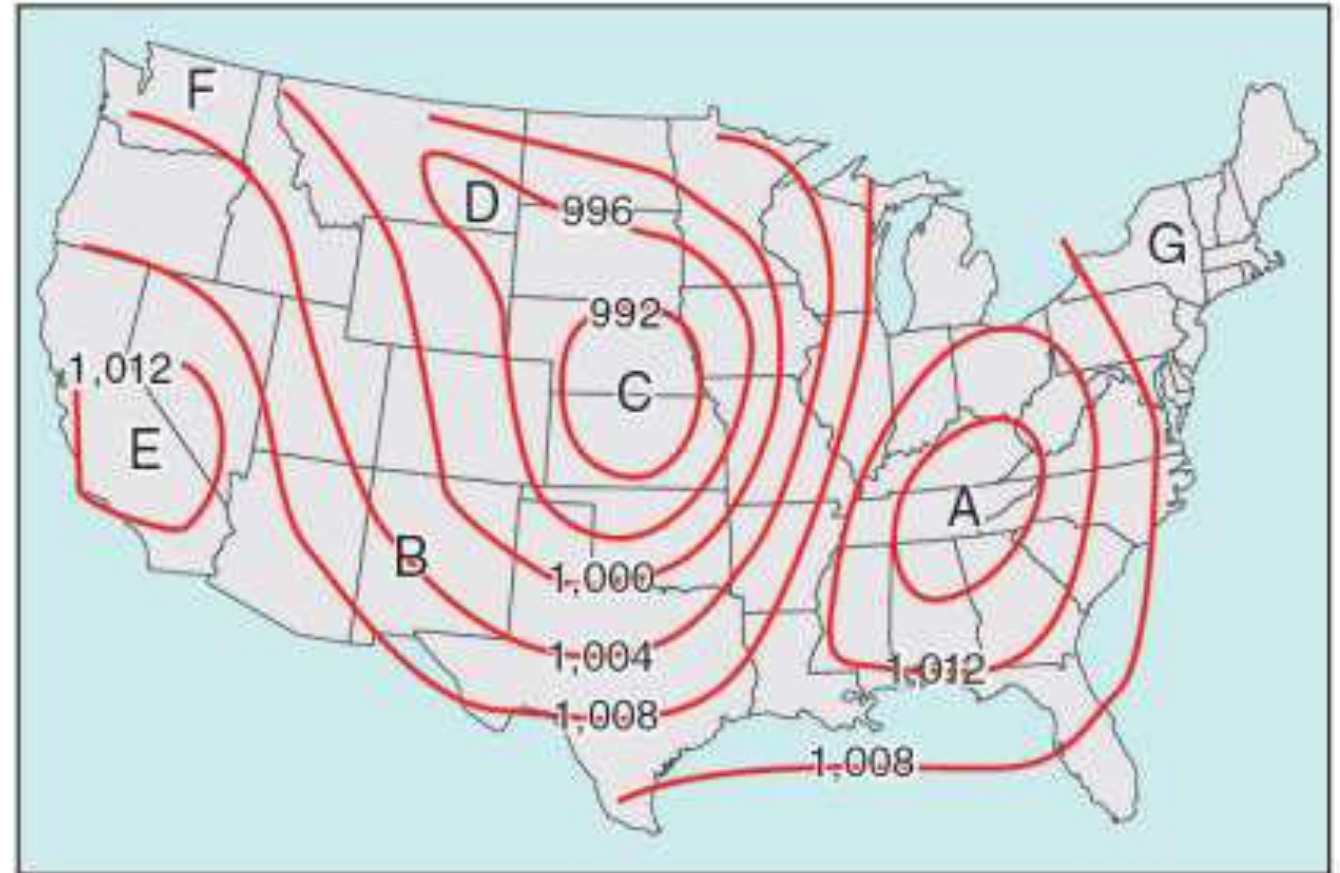
- Calculate the highest-pressure difference observed from the isobars.

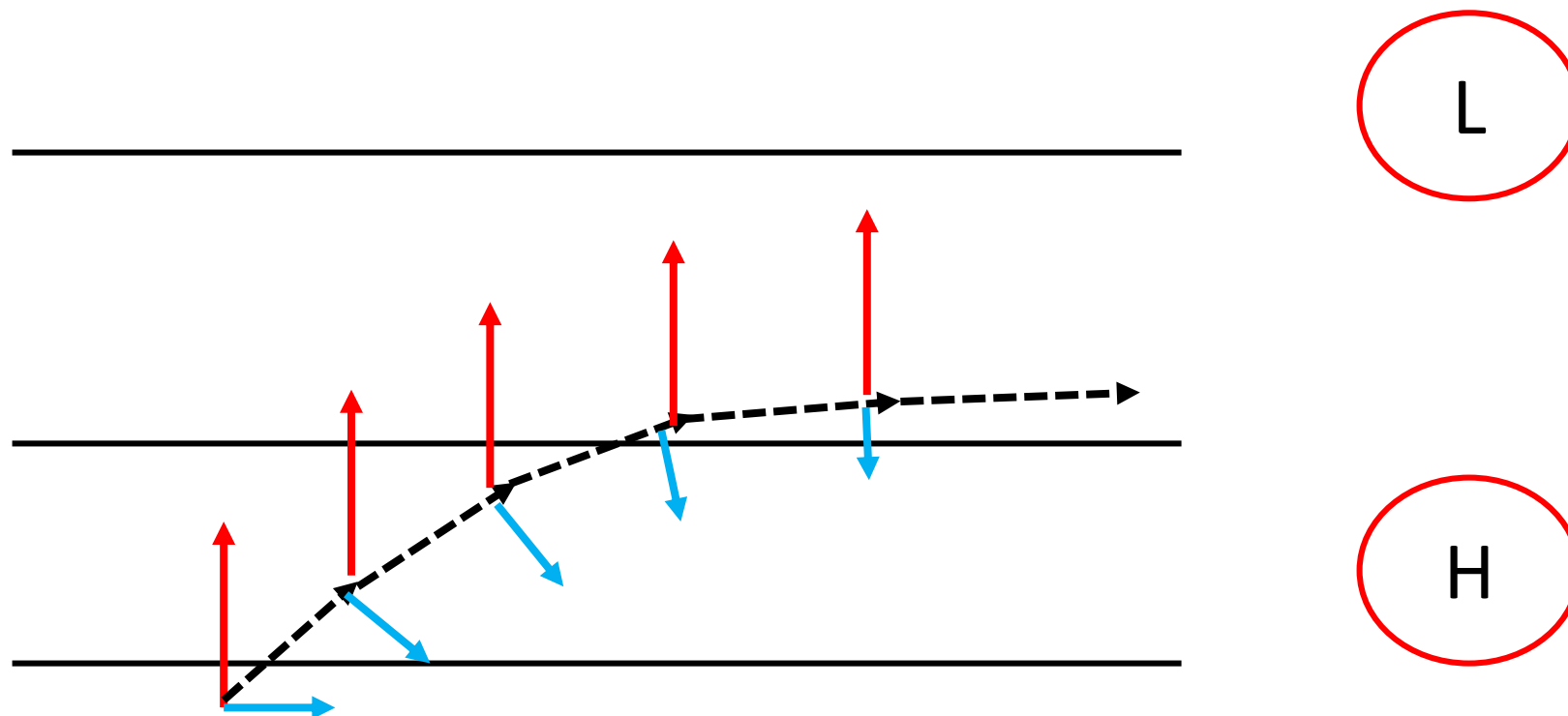


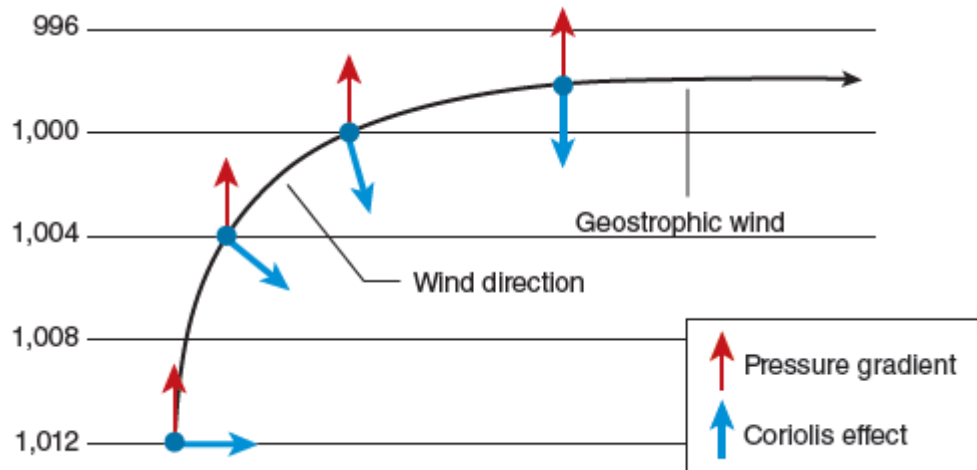
Example Question – 2

Answer the following questions based on the figure given below:

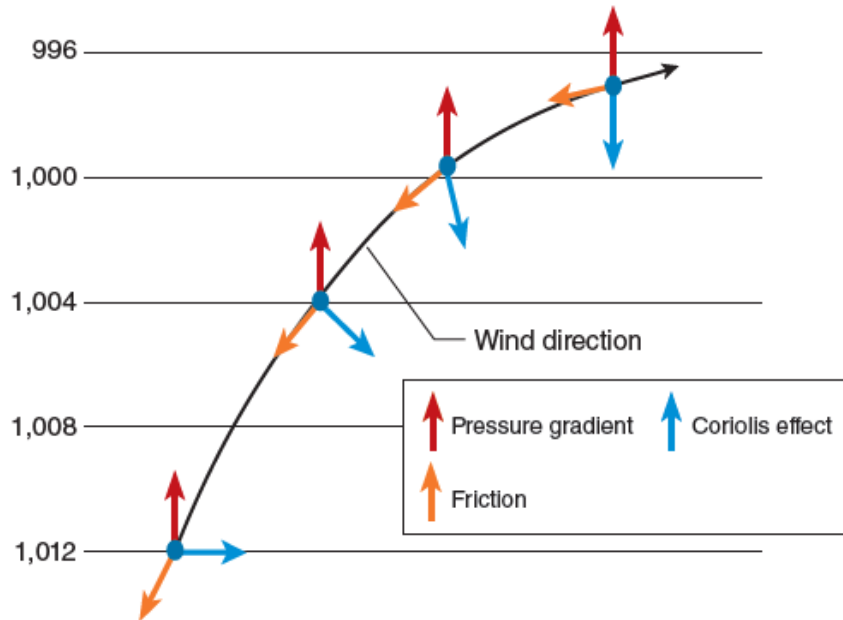
1. Which point on the map is at the center of a low-pressure system?
2. If Earth did not rotate, wind would blow directly from _____.
3. Where is the biggest pressure gradient?
4. Where will you see clear skies ?



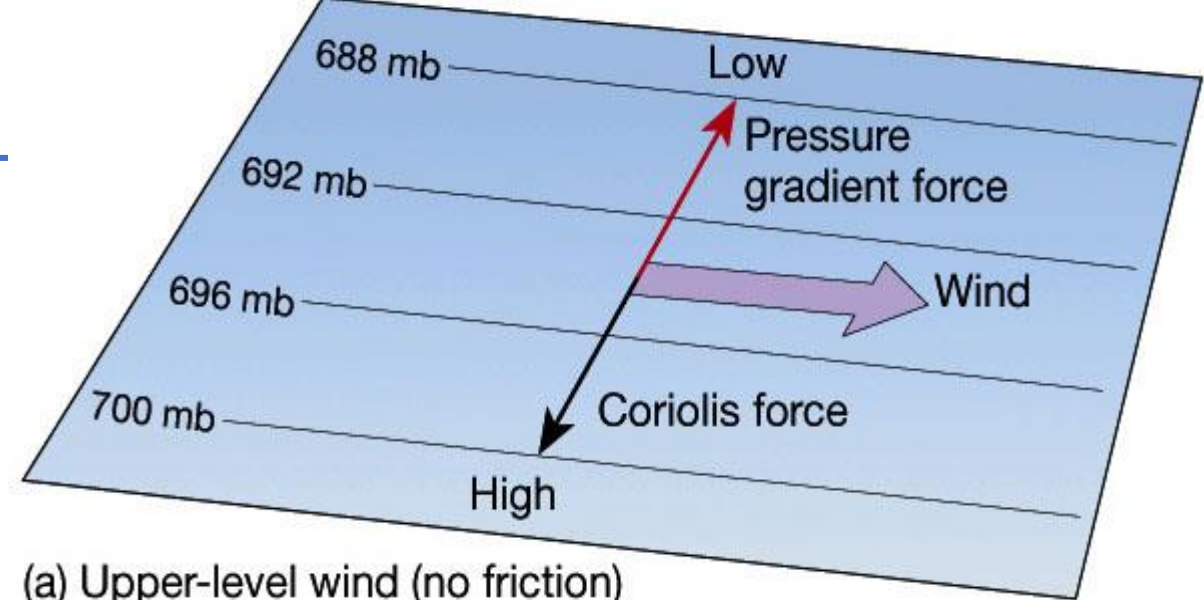




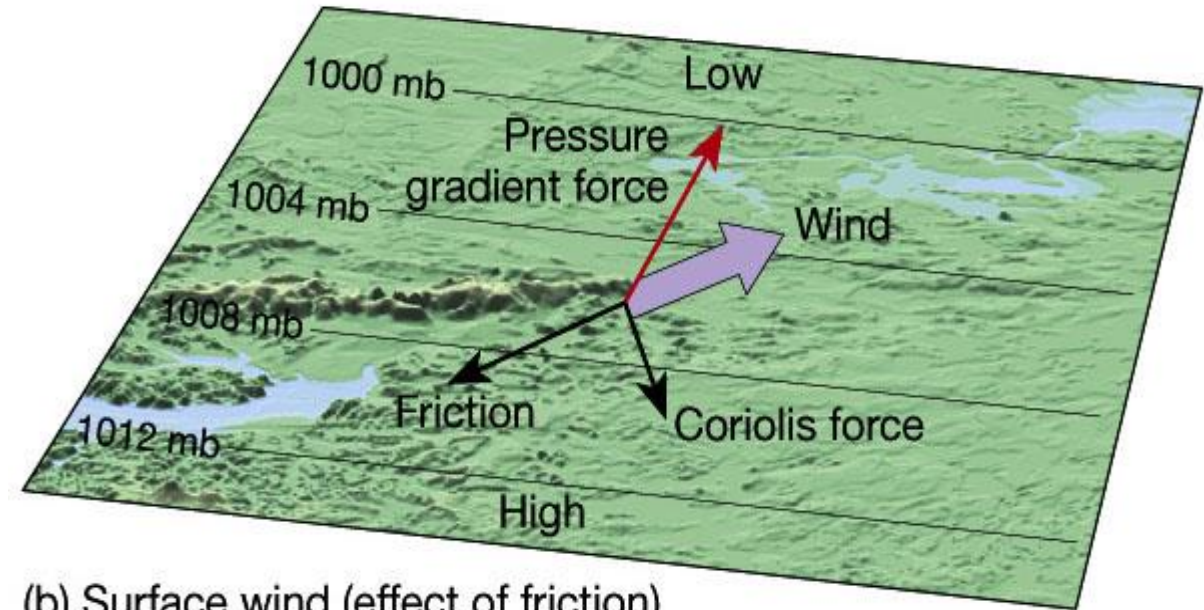
The pressure gradient causes winds to flow from high pressure toward low pressure. Eventually, the pressure gradient is exactly offset by the Coriolis effect, forming geostrophic winds in the middle to upper troposphere that blow parallel to isobars.



Influence of friction. Friction close to Earth's surface reduces pressure gradient and Coriolis effects, resulting in winds that are oriented at an angle to isobars.

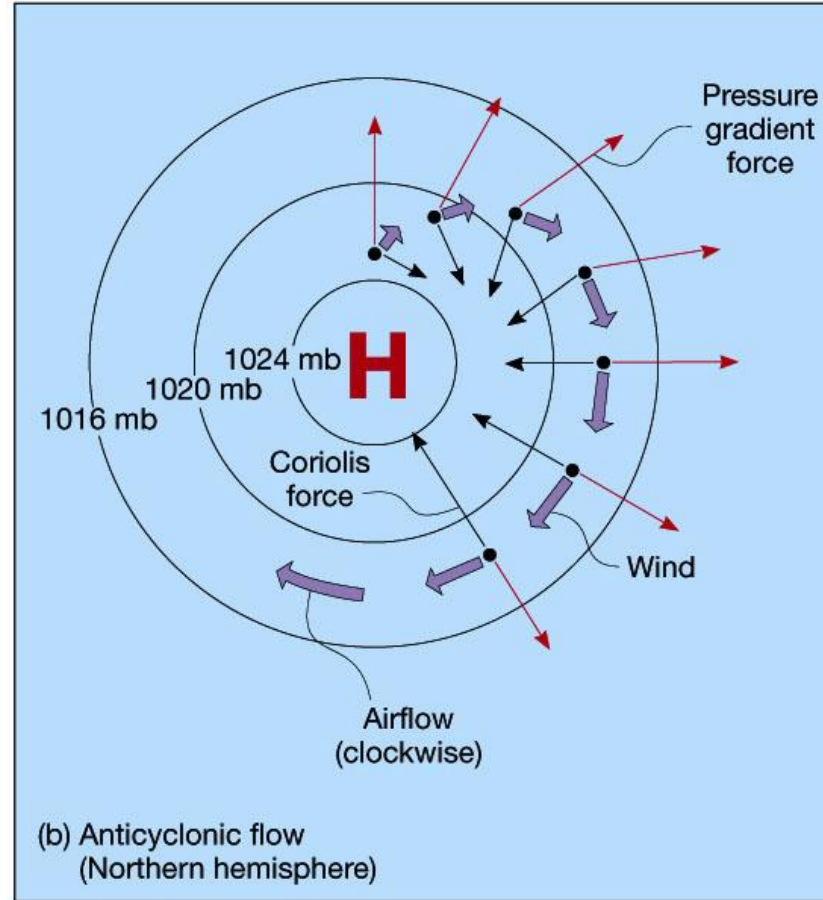
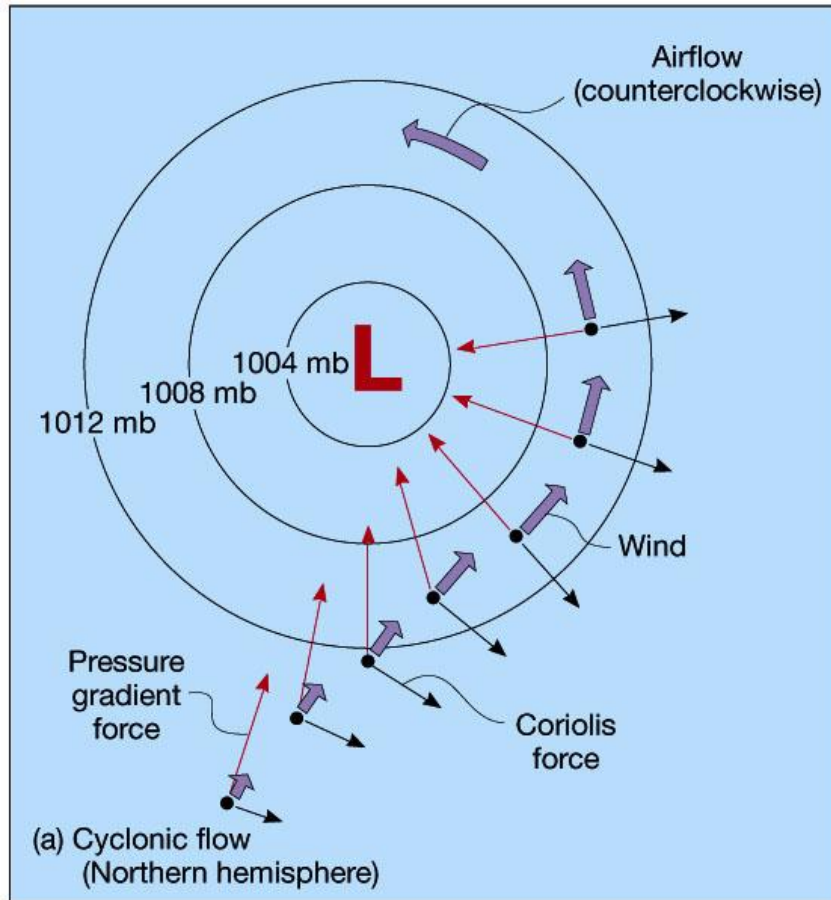


(a) Upper-level wind (no friction)

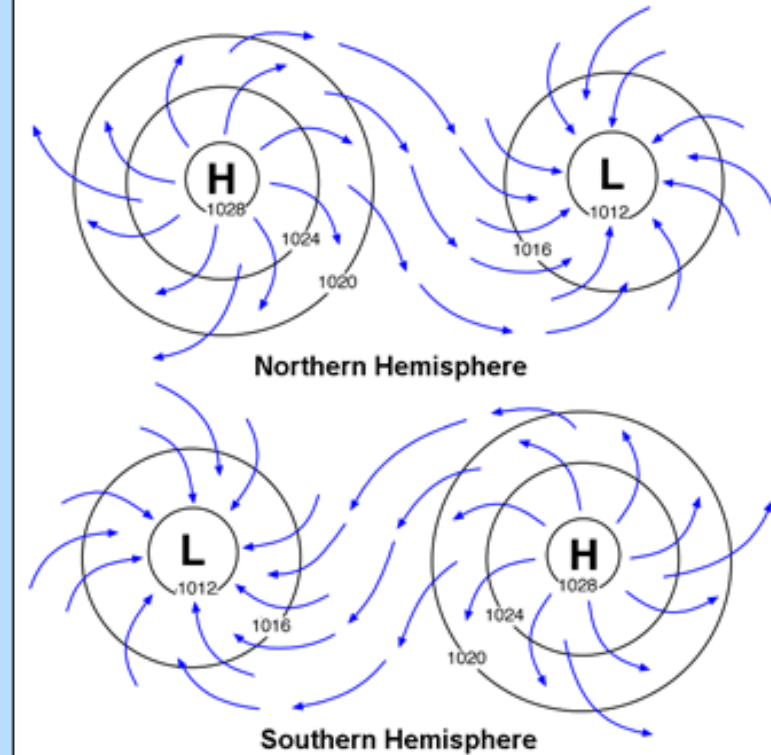


(b) Surface wind (effect of friction)

Winds: Cyclone and anticyclone



High in Atmosphere

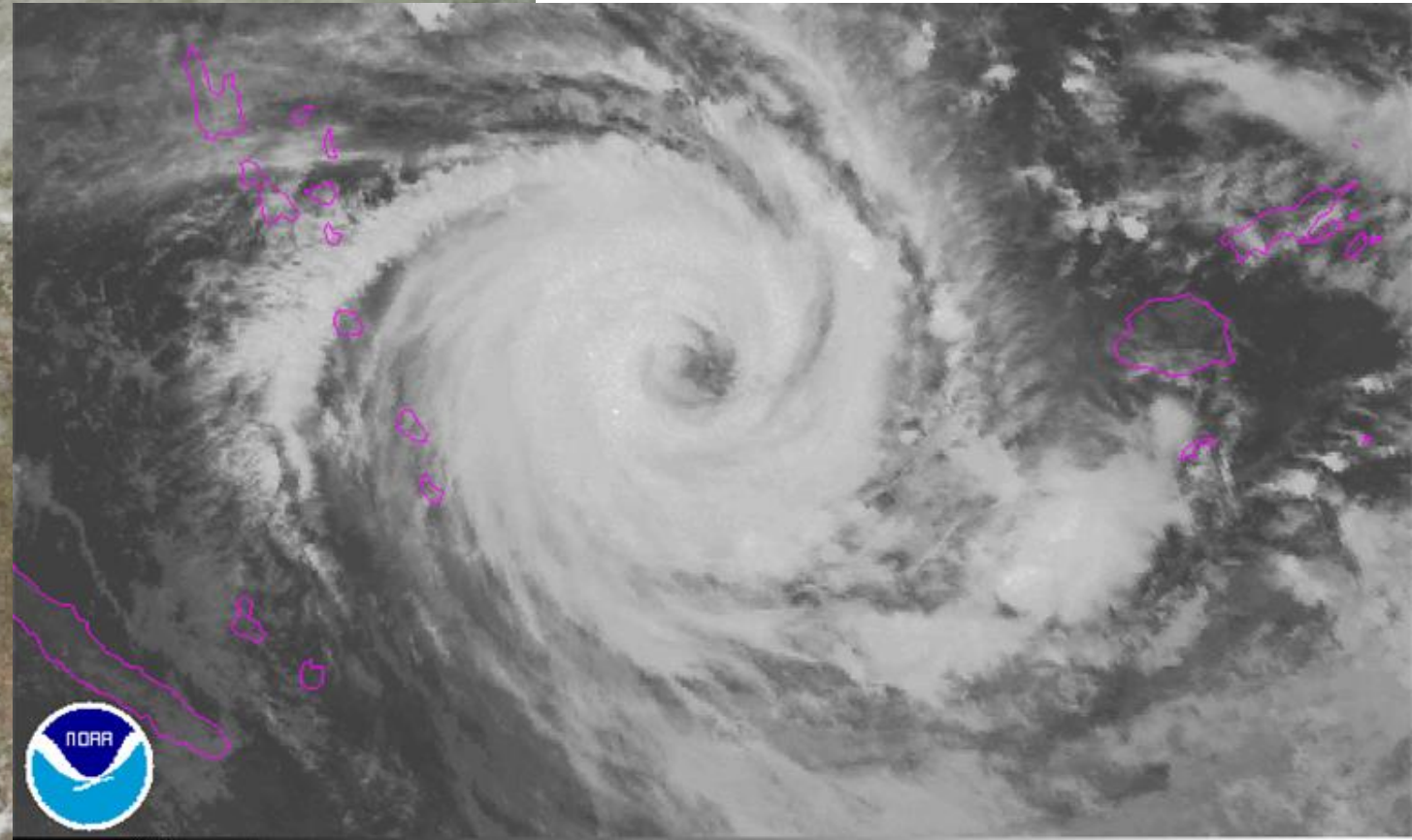
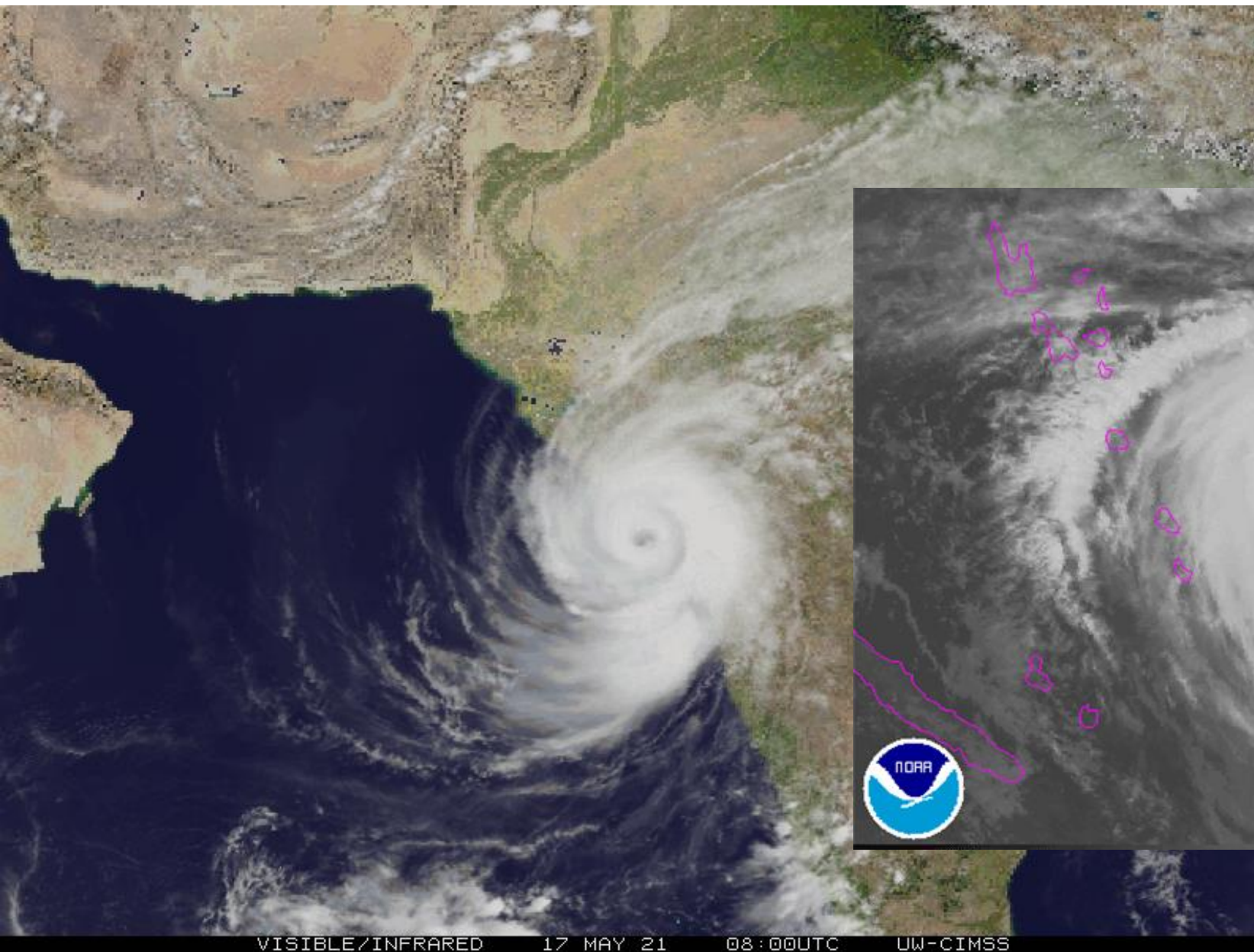


Near surface

Winds

- **The Coriolis effect** influences all moving objects, especially ones moving over large distances (such as intercontinental).
- The Coriolis effect causes objects or moving masses of air to:
 - Change direction—not speed.
 - Maximum Coriolis effect occurs at poles.
 - No Coriolis effect occurs at equator.
- **Rotation of pressure systems due to the Coriolis effect:**
- **Northern Hemisphere:**
 - High pressure turns clockwise
 - Low pressure turns counter-clockwise
- **Southern Hemisphere: opposite of N.H.**
 - High pressure turns counter-clockwise
 - Low pressure turns clockwise

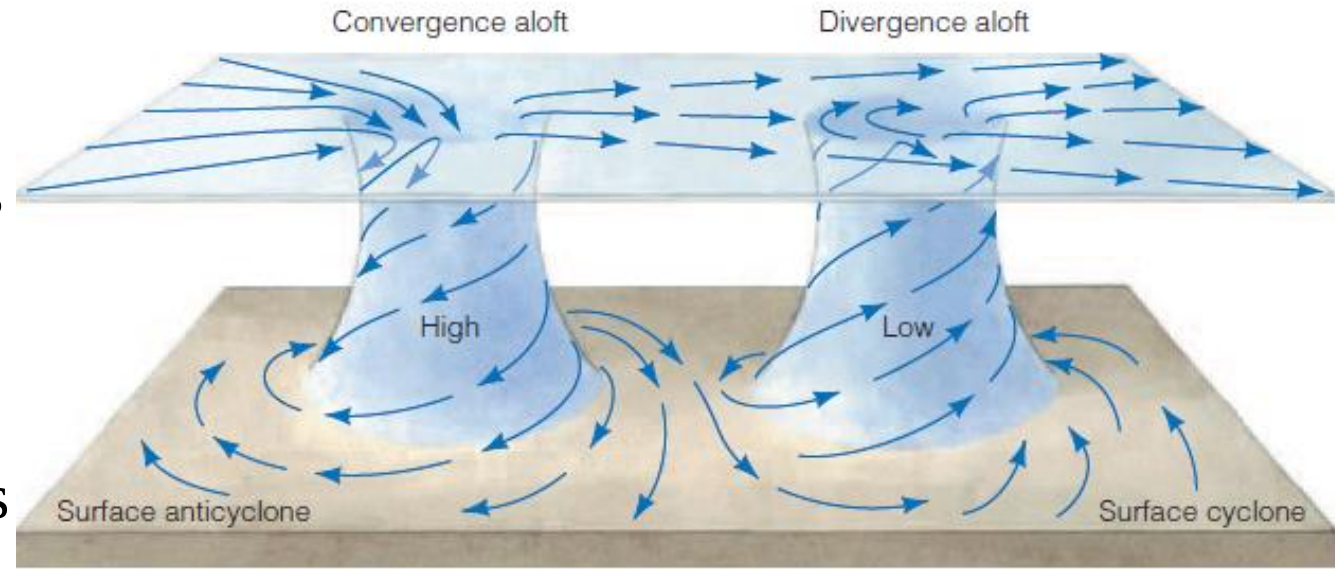




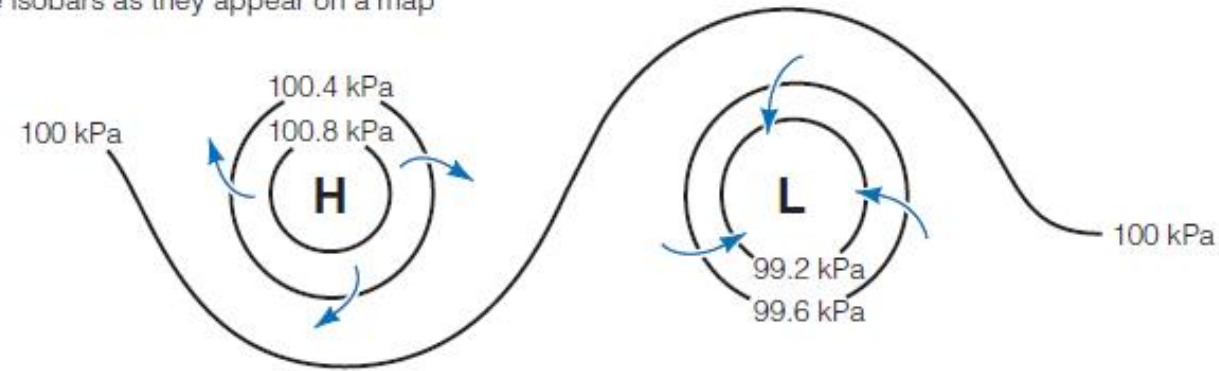
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Winds: Convergent and Divergent Flow

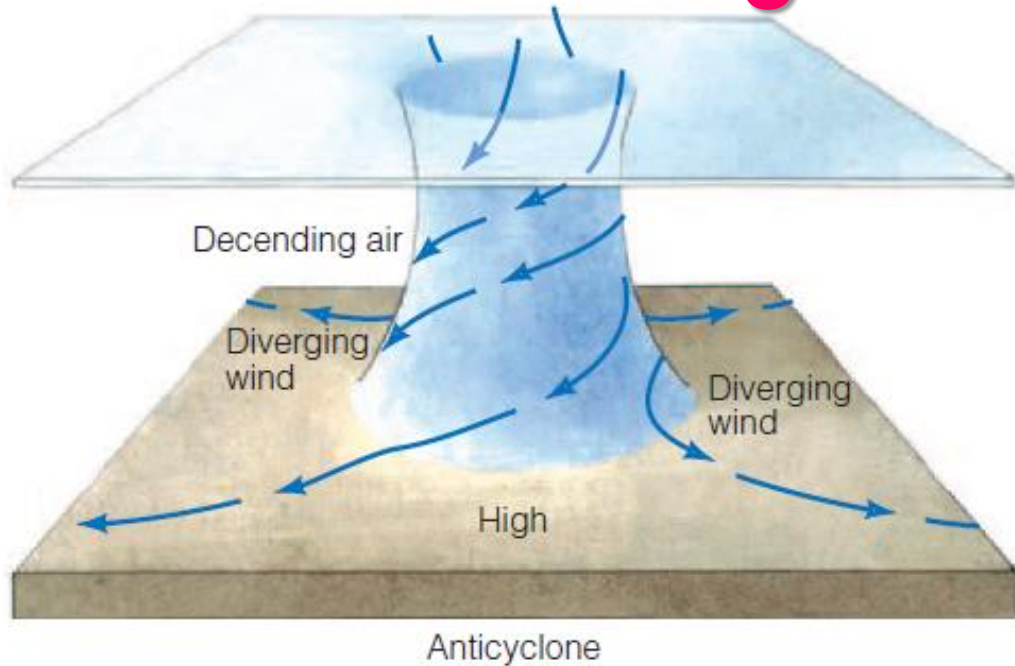
- Airflow spirals outward from a high-pressure area.
- The inward spiral flow in a cyclone causes convergence, which leads to an upward flow of air at the center of the low.
- The outward spiral flow in an anticyclone, in contrast, causes divergence, which leads to an outward flow of air from the center. This outward flow means that high-altitude air will be drawn downward into the center.
- Cold air that is drawn downward will be compressed and heated adiabatically, causing the relative humidity to decrease and leading to clear, cloudless skies.



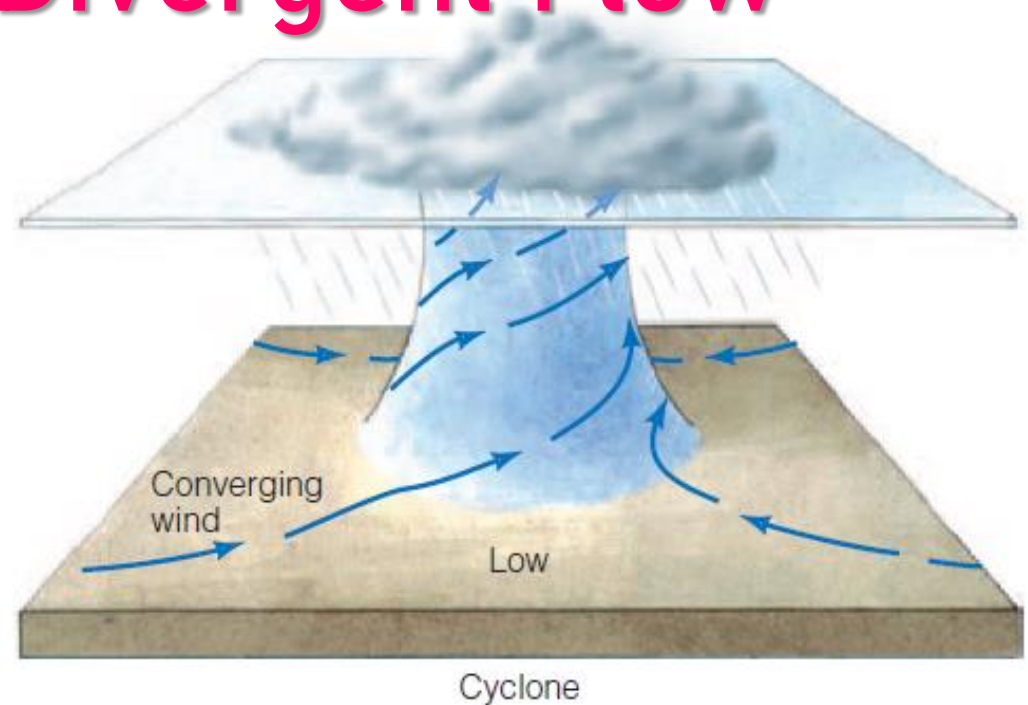
Surface isobars as they appear on a map



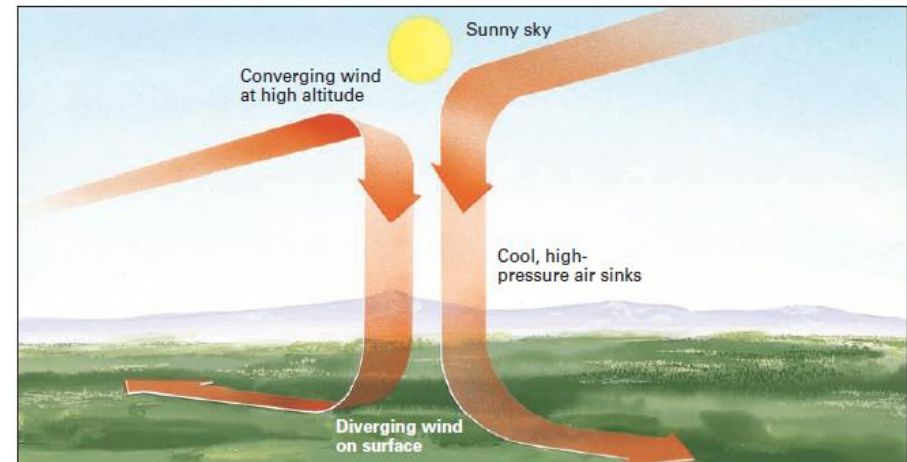
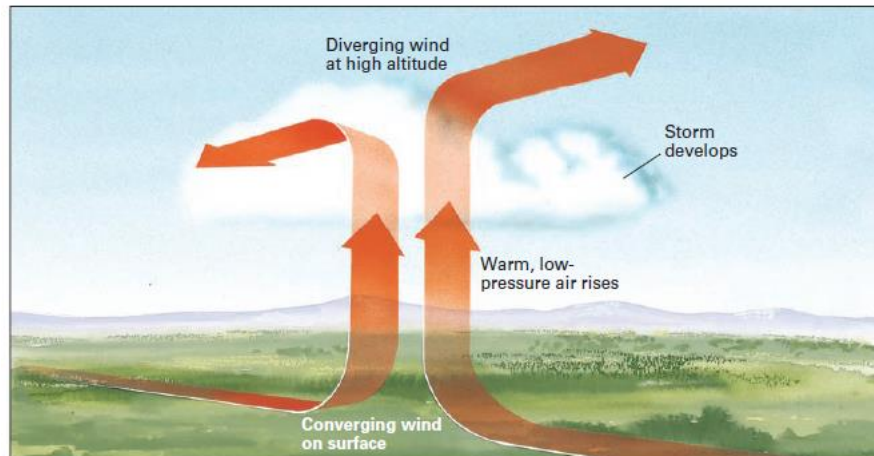
Convergent and Divergent Flow



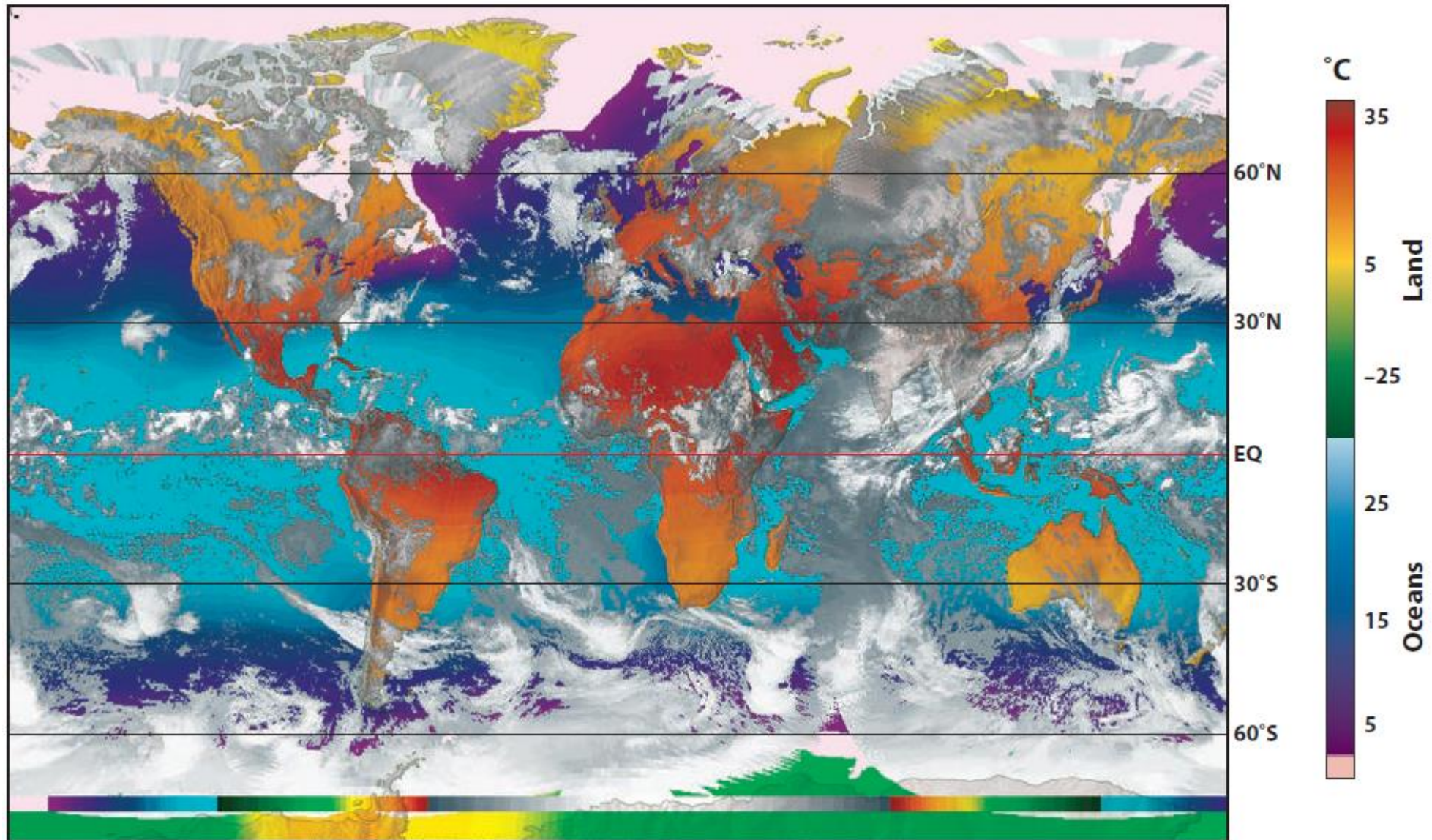
Divergence in an anticyclone draws in dry, high-altitude air, clear skies and fair weather are the result



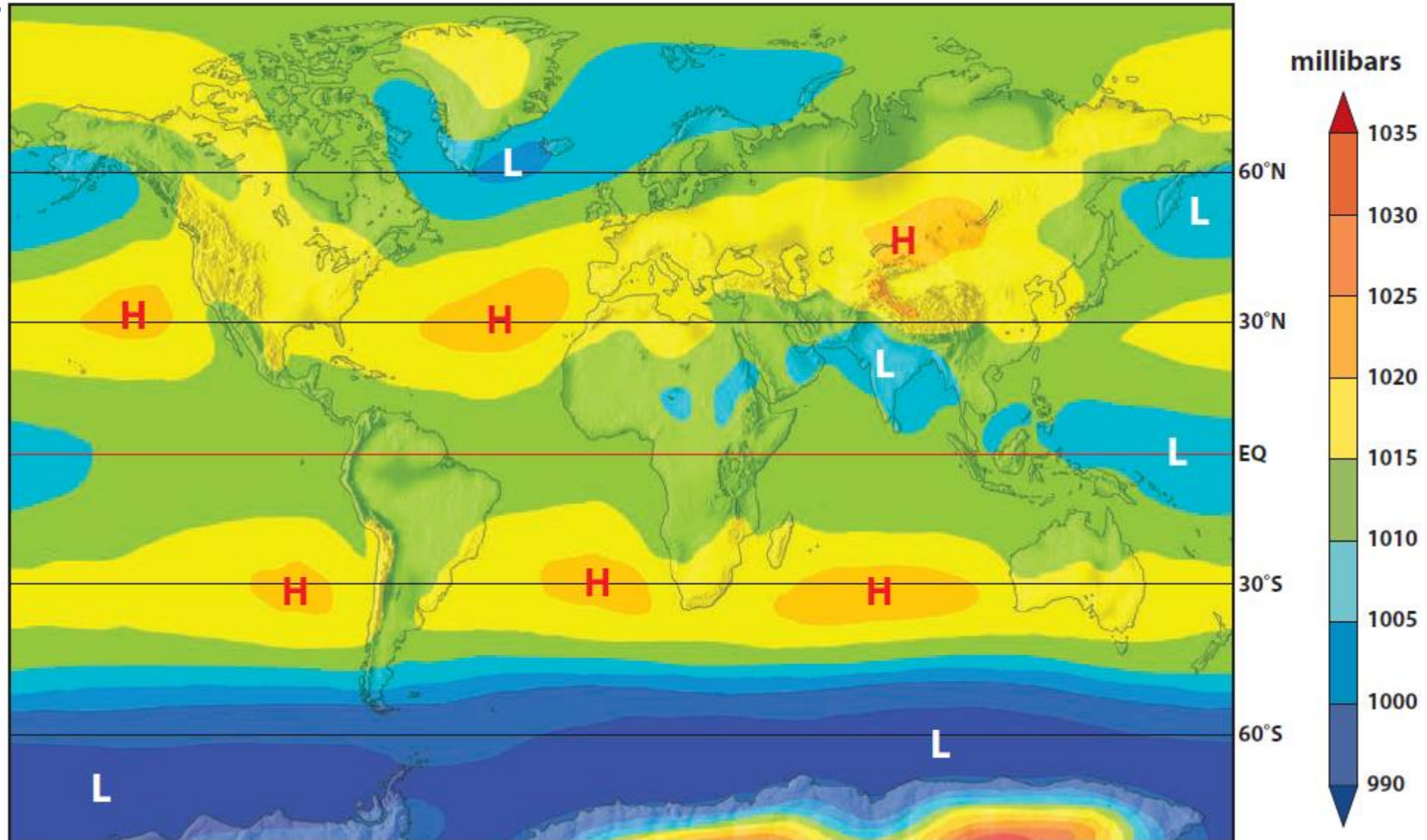
Convergence in a cyclone causes a rising updraft of air and with it clouds and probably precipitation



Temperature and Cloud Cover

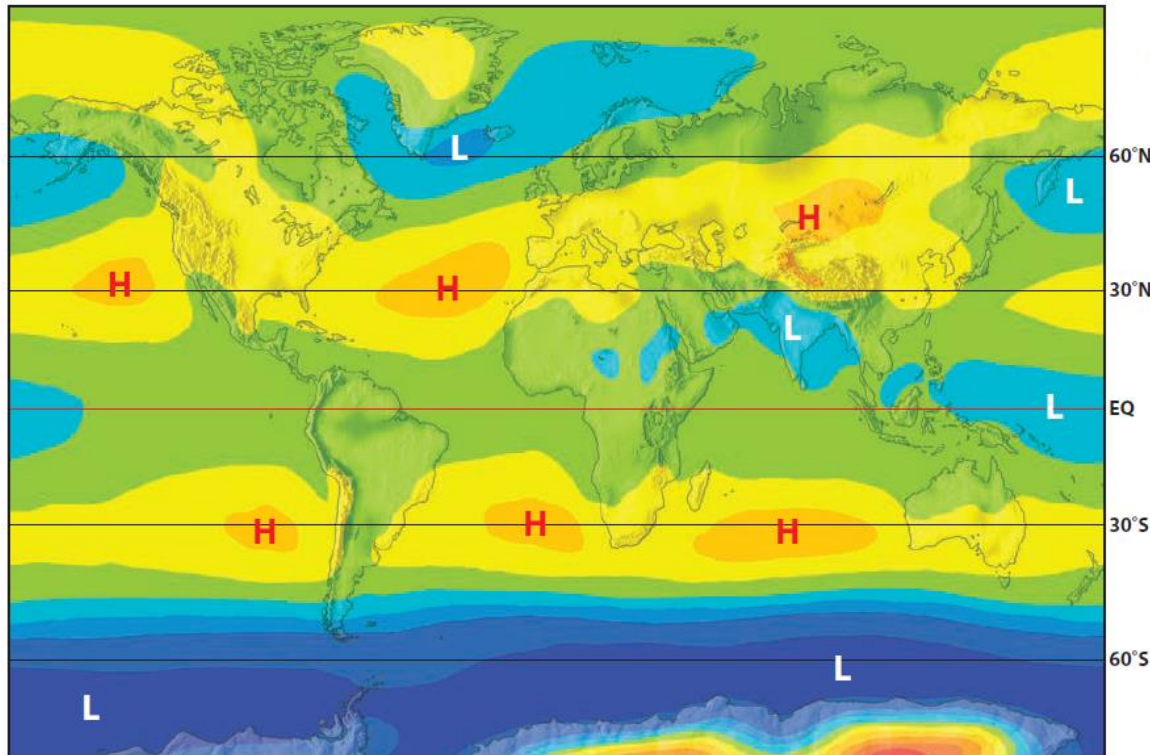


Air Pressure

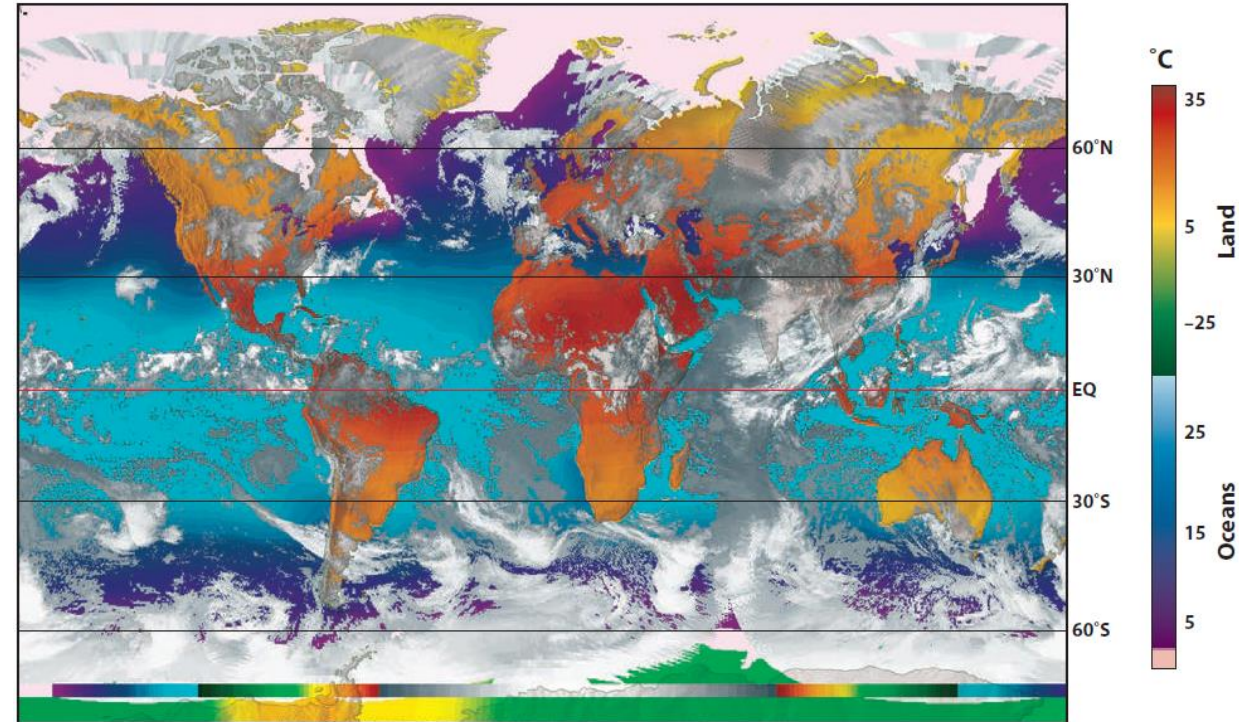
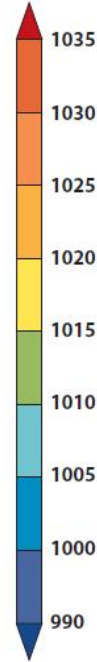


Annual mean sea level atmospheric pressure (millibars) averaged from 1979–1995.
L = low pressure center, H = high pressure center.

Pressure map vs Cloud cover map



millibars



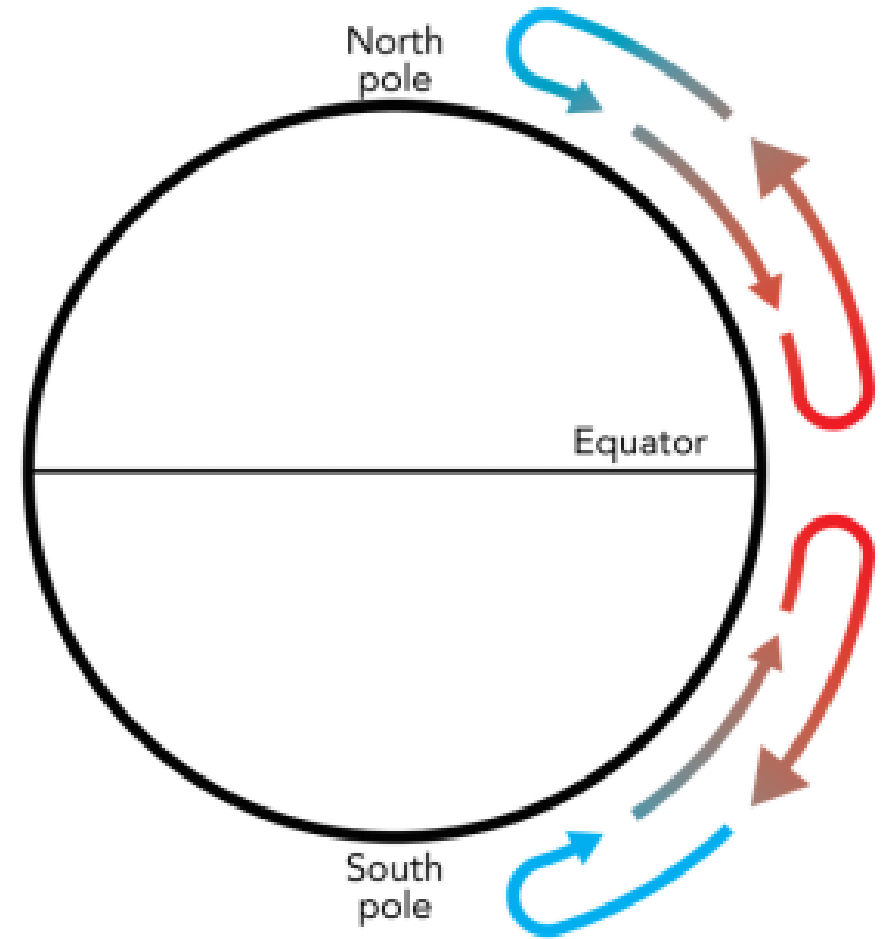
Global atmospheric circulation

- **Why temperatures decrease with increasing latitude as one moves north and south from the equator ?**
- Solar radiation (sunlight) strikes Earth more directly at the equator and the tropics than at the poles. Locations closer to the equator receive about 2.5 times more solar energy than locations at or near the poles.
- **Which will have highest and lowest temperature range – oceans or continental crust ?**
- The temperature range (highest to lowest) is much greater for the continents (approximately 60 °C) than for the oceans (approximately 30 °C).
- **What was observed in Cloud cover map ?**
- **Clouds are concentrated in irregular bands parallel to the equator and latitudes 60°N and S and that these latitudes are also characterized by higher precipitation rates and low atmospheric pressures.** In contrast, regions around latitude 30°N and S have fair skies with little cloud cover, higher air pressures, and low rainfall rates; these regions also correspond with the locations of almost all the major deserts on Earth



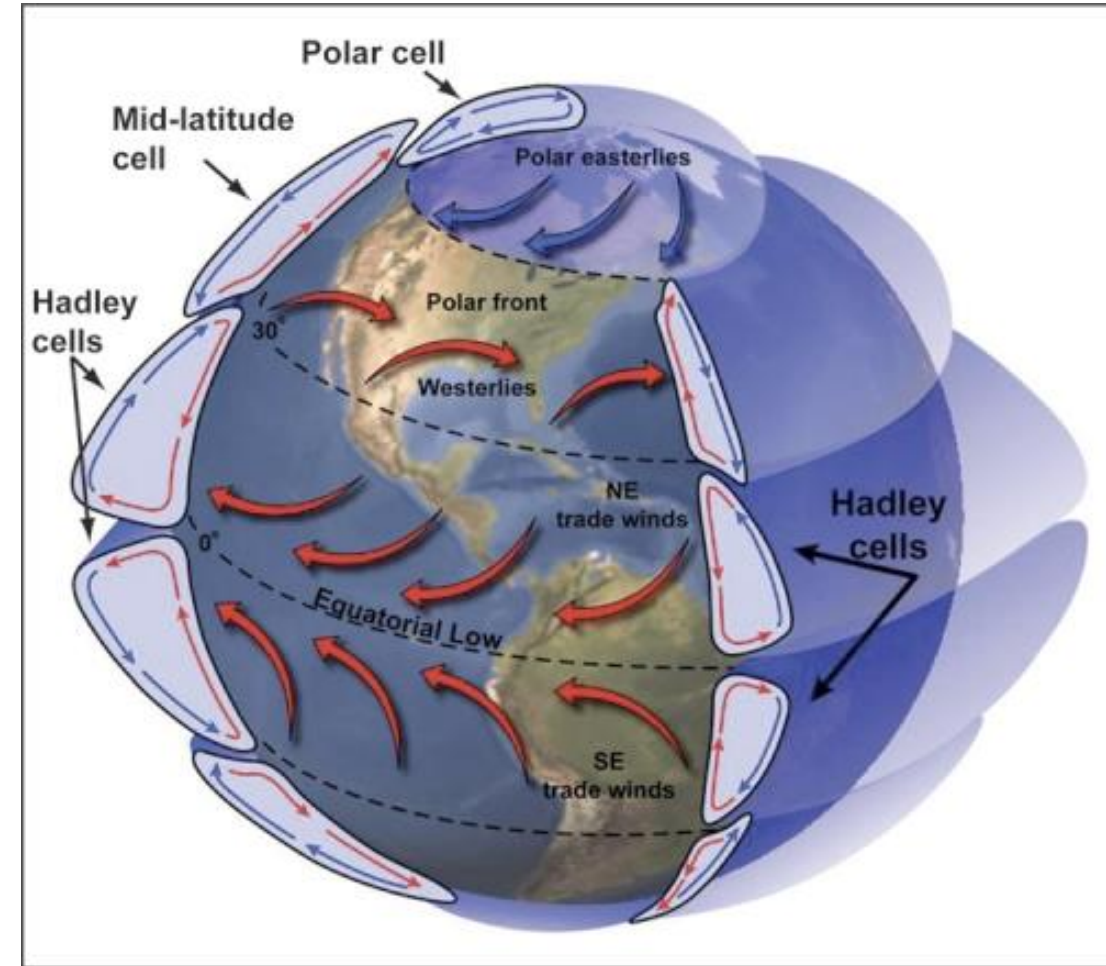
Global atmospheric circulation – Nonrotating Earth Model

- The much warmer air at the equator would rise.
- As the air ascended, it would cool. When it reached the top of the troposphere, it would bump into the base of the warmer stratosphere. Because it could not continue to rise through the stratosphere, the air would have to turn north or south and flow toward the polar regions.
- Far above Earth's surface, this air would arrive at the pole as a mass of cold, dense air. This cold air would sink and move outward toward lower latitudes.
- In the simplest case, **the rising air at the equator and the sinking air at the poles would form opposing limbs (loops) of a large-scale circulation pattern called a *convection cell*.**



Global atmospheric circulation – Rotating earth

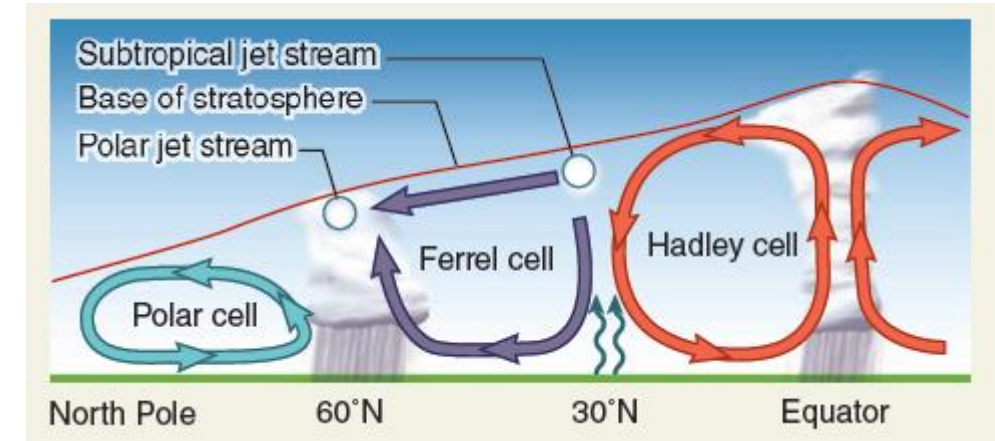
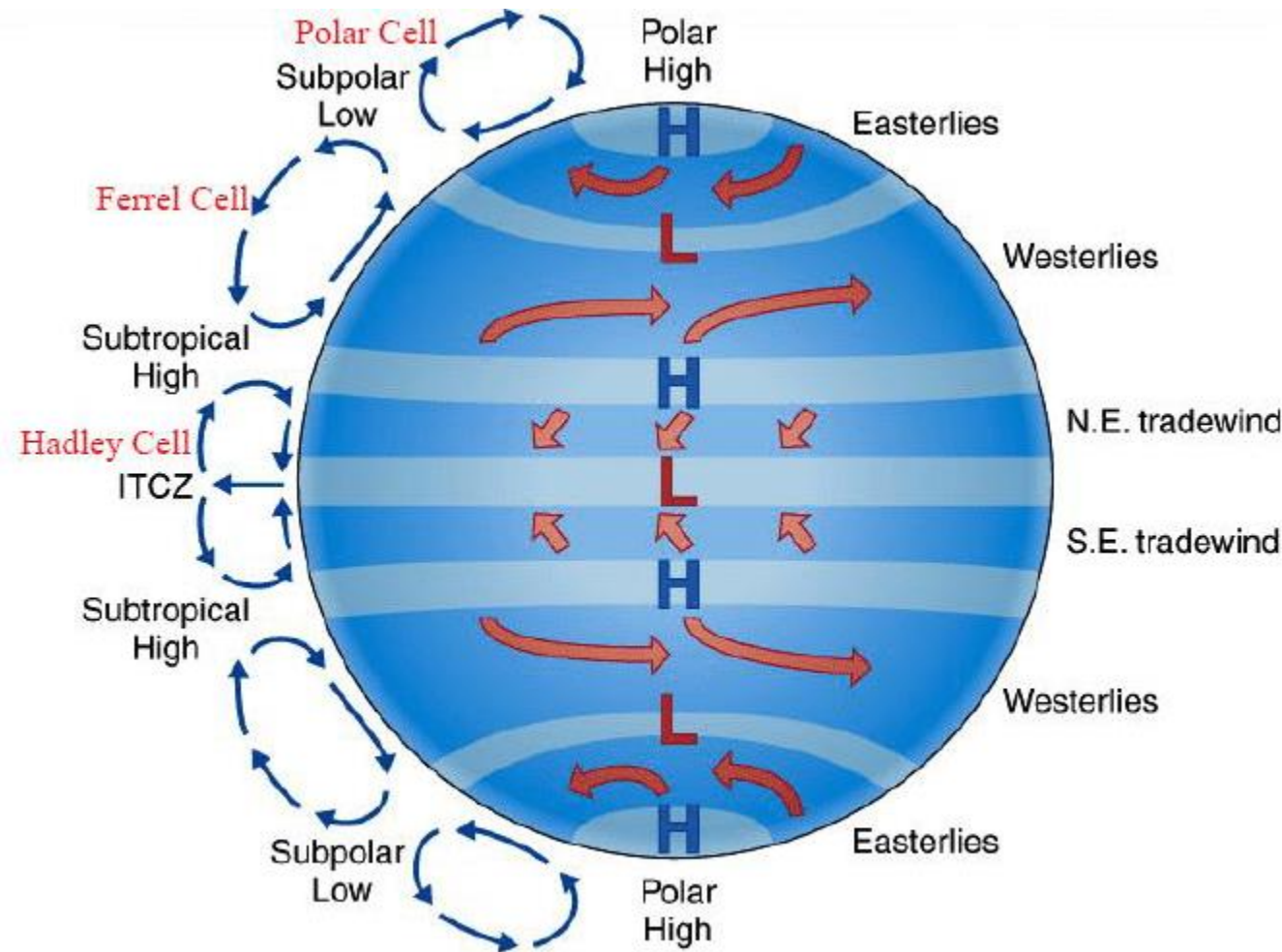
- Circulation Cells in Earth's Atmosphere
- Three major **circulation cells** move air, heat, and moisture through the atmosphere between the equatorial regions to the polar regions.
- These cells are constantly changing due to regional air pressure changes under the influence of the Coriolis effect.
- **Hadley cells** (0° to 30° N and S of equator)
 - Responsible for the **Trade Winds**: They blow NE in N. Hemisphere and SE in S. Hemisphere.
- **Ferrel cells** (30° to 60° N and S of equator)
 - Responsible for the **Prevailing Westerlies** in both hemispheres.
- **Polar cells** (60° to 90° N and S)
 - Responsible for the **Polar Easterlies** in both hemispheres.



Coriolis force actually breaks circulation into three separate convection cells in each hemisphere (6 in total)

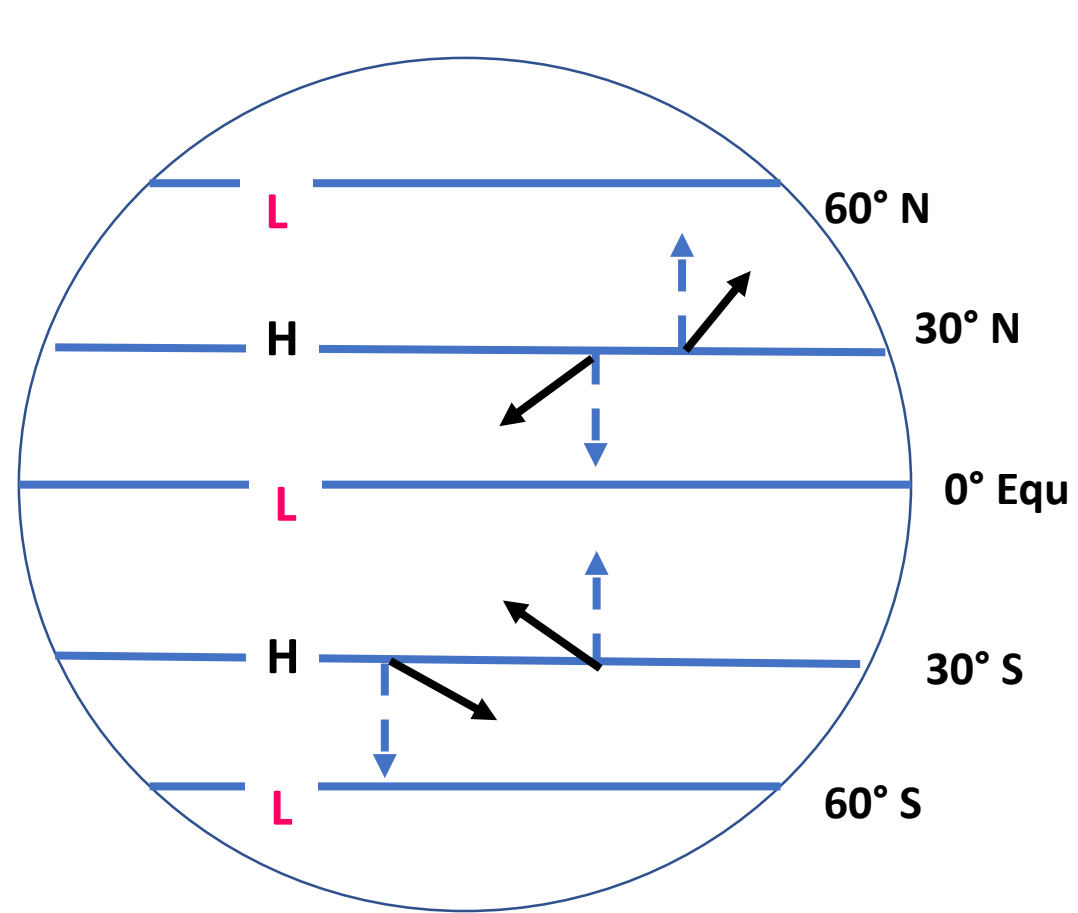


Global atmospheric circulation – Rotating earth

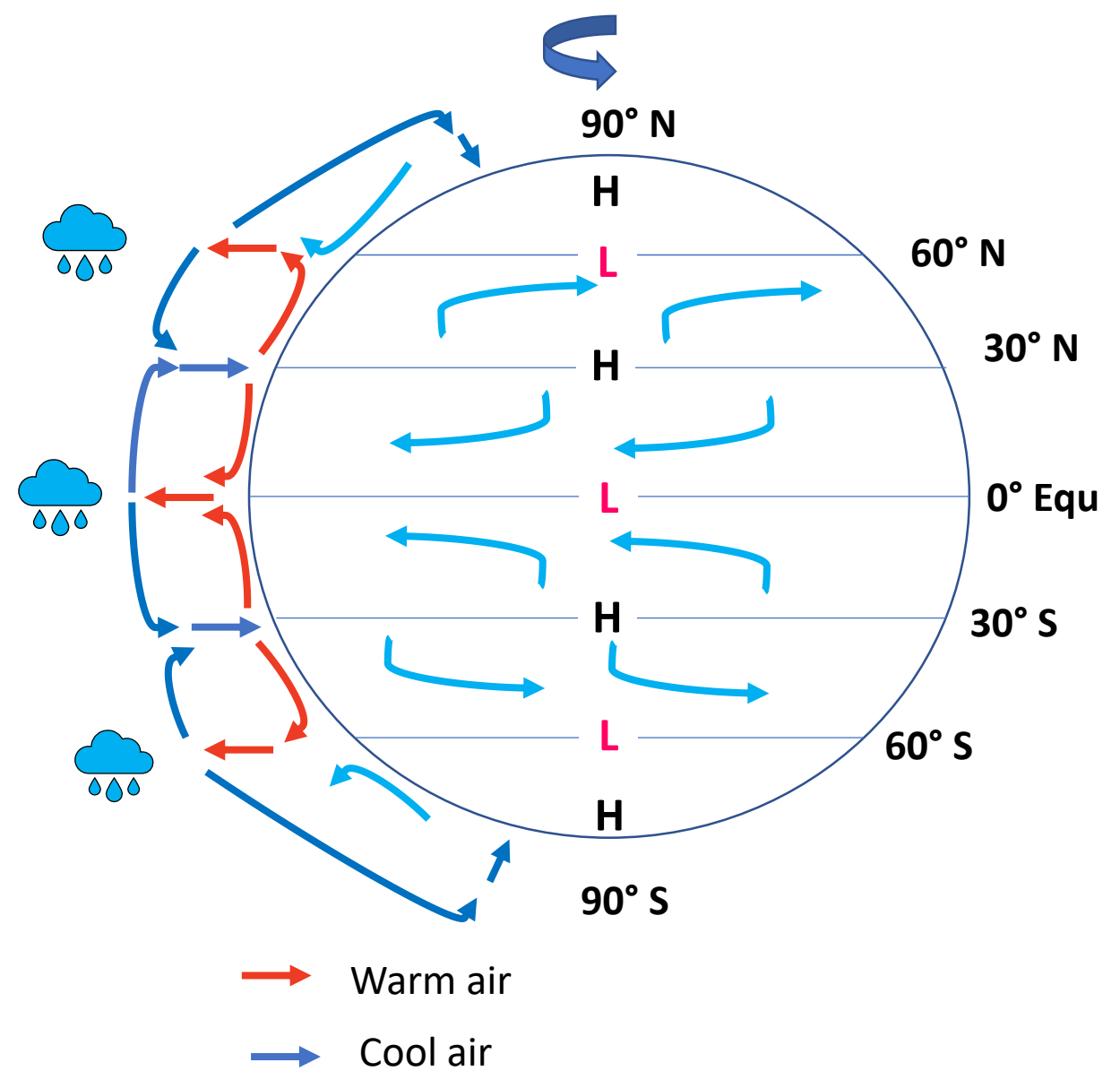


- Thermally indirect cell – driven by eddies (weather systems), not thermal energy.
 - Ferrel cell
- Thermally direct cell – driven by thermal energy, air rises in warm area, air sinks over colder area.
 - Hadley cell
 - Polar cell

Inter-tropical convergence zone (ITCZ) the region that circles the Earth near the equator, where the trade winds of the Northern and Southern Hemispheres converge



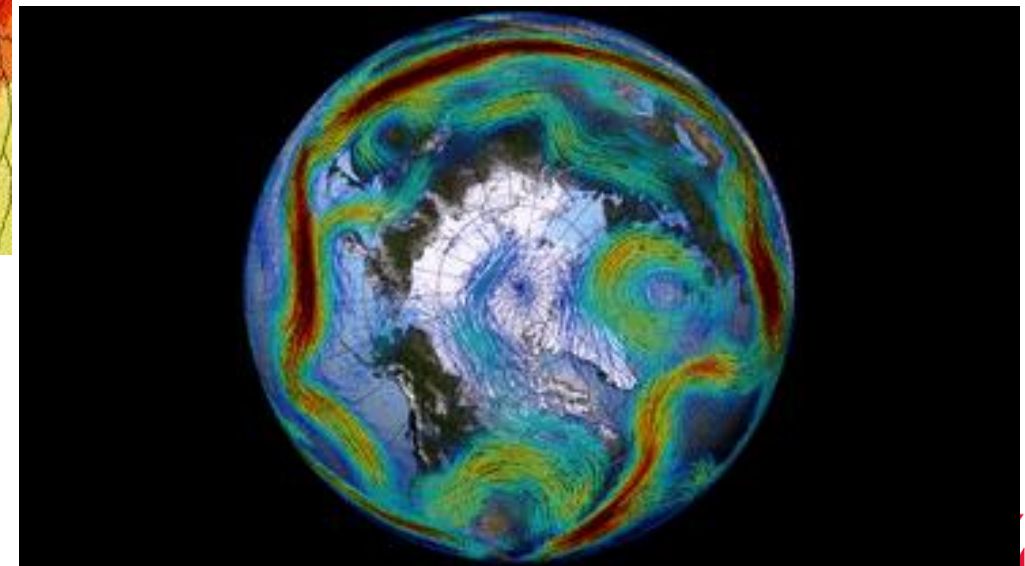
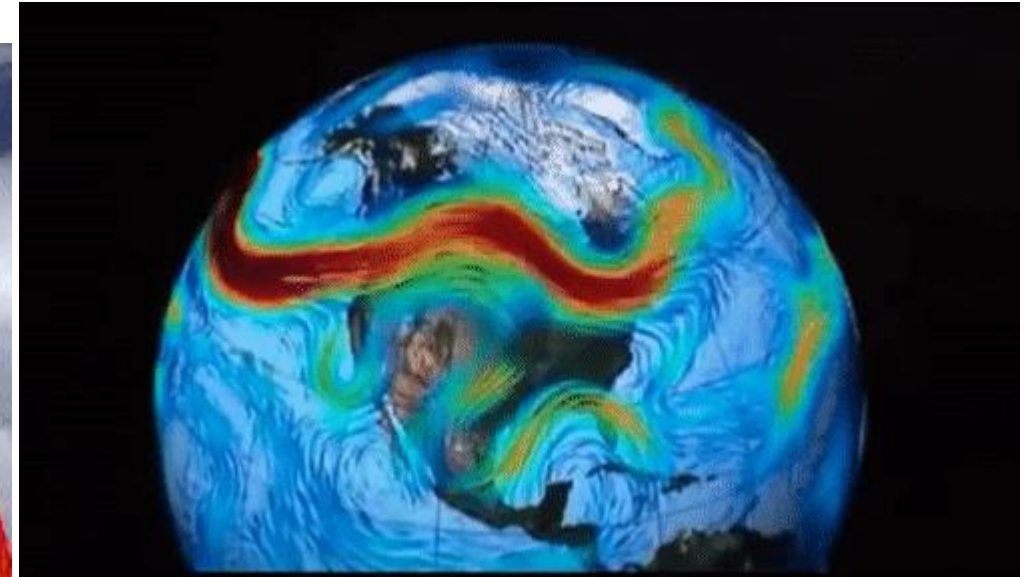
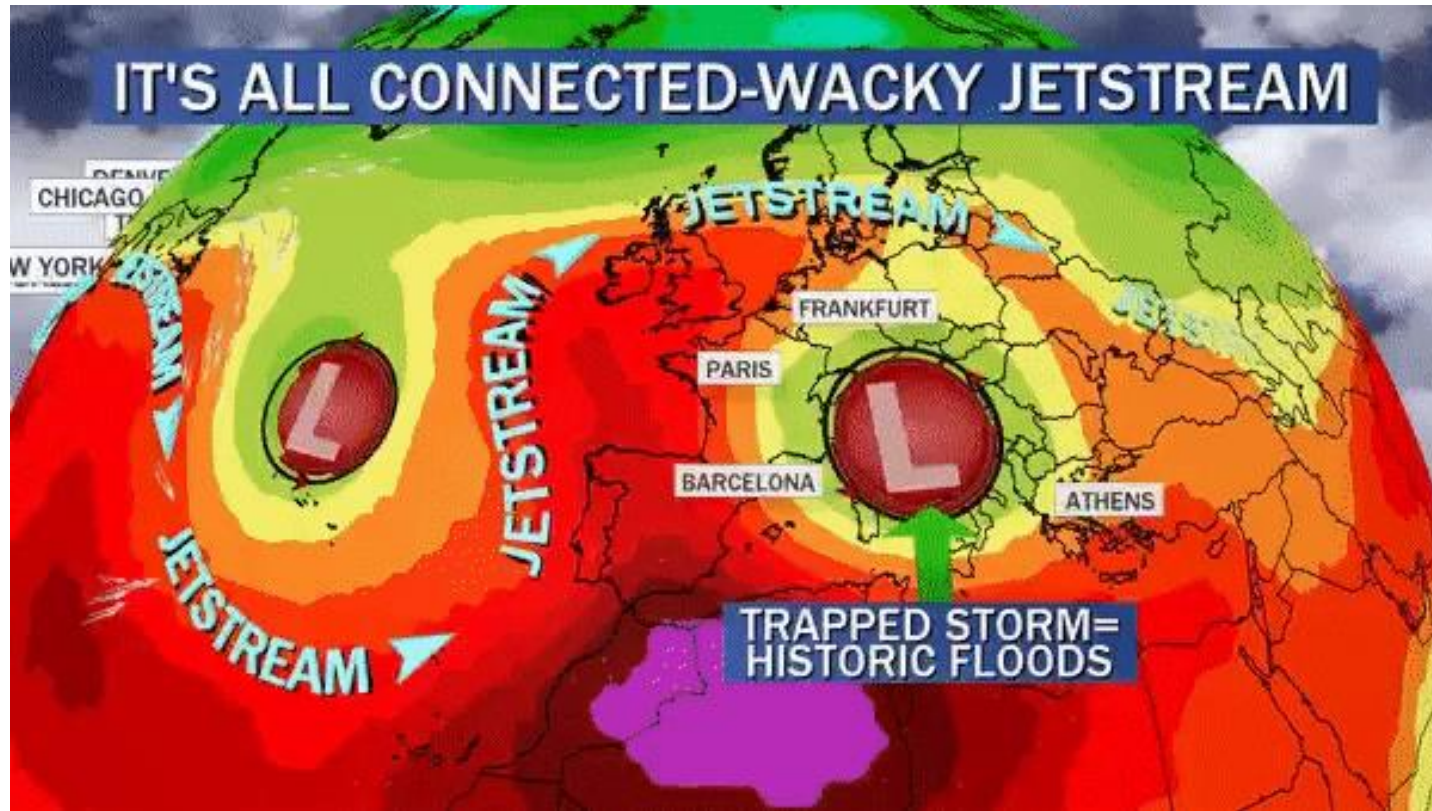
- - - - - Wind direction with no Coriolis effect
 ————— Wind direction with Coriolis effect



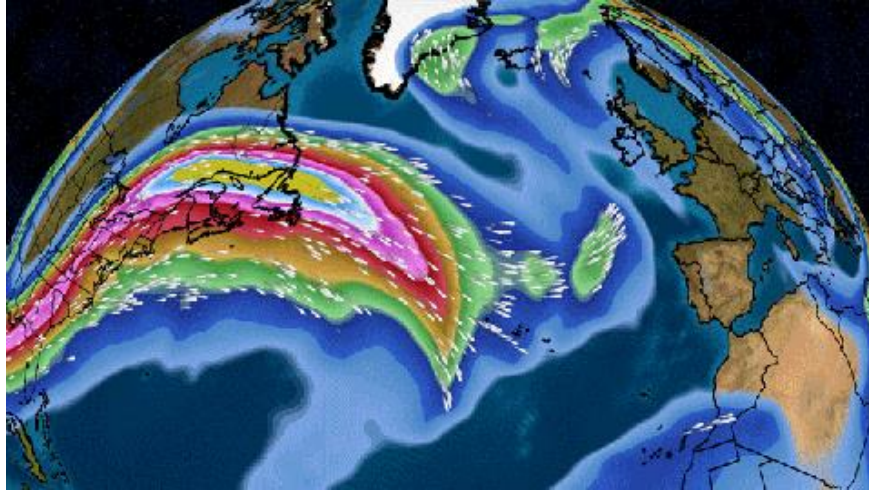
————— Warm air
 ————— Cool air

The regions where warm air rises are more likely to experience increased cloud cover and rainfall. This is because when warm air rises, it cools and condenses, forming clouds and eventually leading to precipitation.

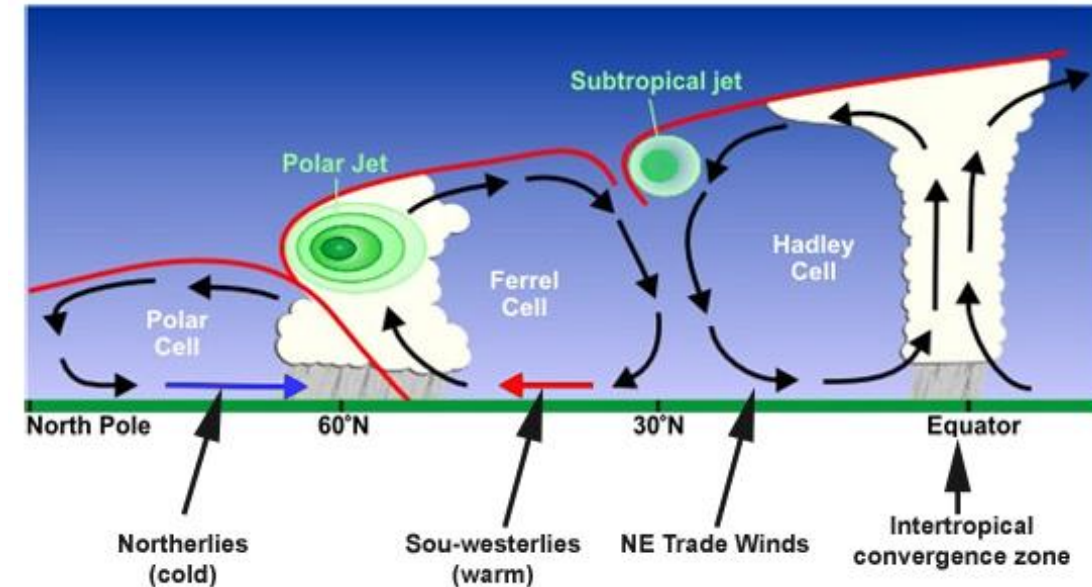
Global atmospheric circulation – Jet Stream



Jet Stream



- A **jet stream** is a narrow, variable band of very strong winds in the upper troposphere.
- They are predominantly **westerly air currents** encircling the globe several miles above the Earth.
- There are typically two or three jet streams in each of the northern and southern hemispheres. These high-speed wind currents often move at speeds exceeding 250 miles (400 km) per hour at altitudes of 6 to 9 miles (10 to 15 km).
- Jet streams are influenced by moving air masses and the Coriolis effect causing them to meander and sometime split.



How does climate change affect Jet stream ?

- If the planet continues warming at a high rate, the jet stream is likely to shift north over time.
- As this temperature difference decreases, it causes a slight drop in zonal winds in the jet stream — which, in turn, leads to more meandering of the jet stream.

Can we stop Jet stream ?

- Two crucial ingredients responsible for today's jet stream: heating from the sun and the spinning of the Earth. **We can't turn off the jet stream.**

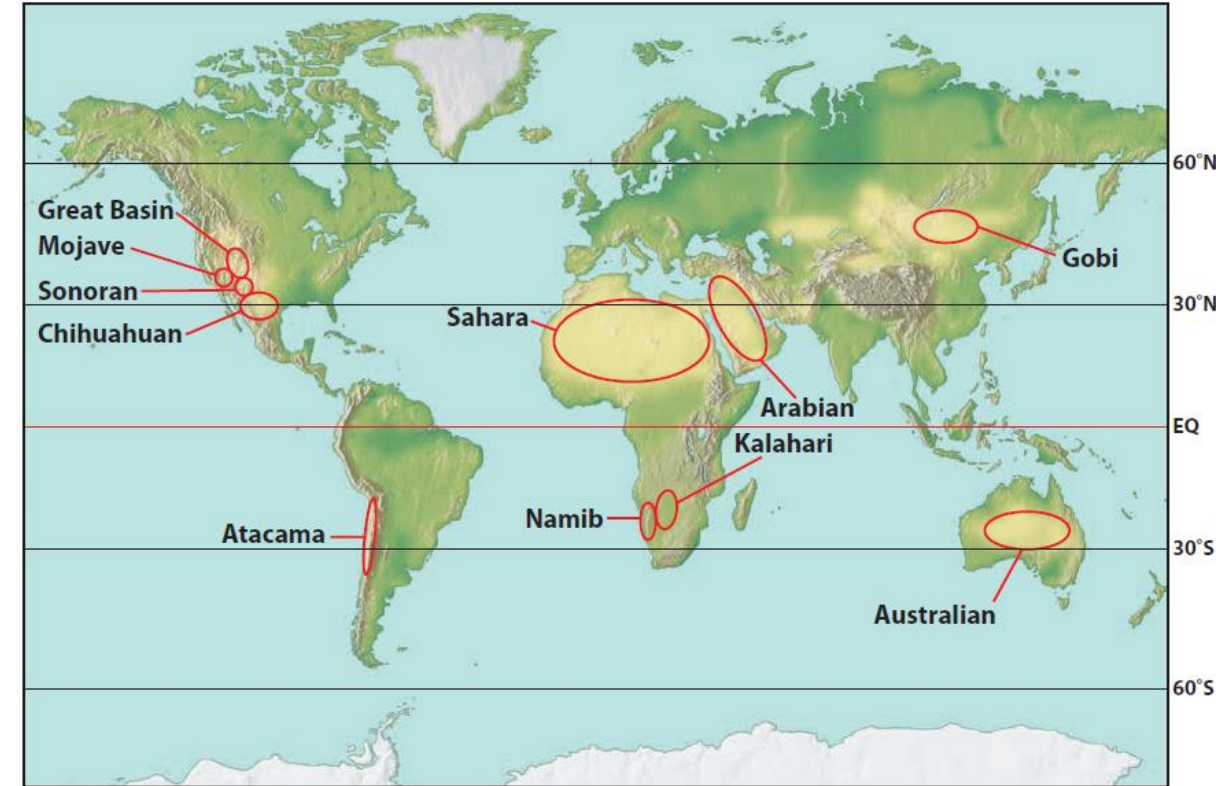


Deserts

Most of the world's hot deserts are located in two belts near the Tropics of Cancer and Capricorn, roughly 15 to 30 degrees north and south of the Equator.

Why They Are Located There

- **Sinking Air:** Warm air rises at the equator, cools, and drops heavy rain in the tropics. This dry air then moves toward the 30-degree latitudes, sinks down, warms up, and stops clouds from forming.
- **Distance from Oceans:** Lands deep inside continents are too far from the sea. Water winds lose all their moisture before they reach these inner areas.
- **Mountain Barriers:** Mountains block rain clouds from passing over. The clouds drop their rain on one side and cross over as dry air to the other side.
- **Cold Ocean Currents:** Cold water near western coasts cools the air above it. This stops rain clouds from forming, creating very dry coastal lands.



Super storms



Super storms

- Large rotating storms are called **hurricanes** (near North America), **typhoons** (near Southeast Asia) and **cyclones** (in the Indian Ocean).
- All are the same, caused by warm moist winds being drawn to the center of low pressure near the center of the storm (called the **eye** in well developed storms).
- North of the equator the Coriolis effect causes low-atmospheric pressure to rotate counterclockwise, but south of the equator they rotate in a clockwise direction.
- The lower the air pressure in the eye of the storm, the greater the wind speed and rotation.
- **Note on the map that there are no hurricanes along the equator or near the poles. These are regions where the Coriolis effect is not a significant force in deflecting storm winds to cause rotation.**
- Superstorms not only can cause major wind damage and flooding, but can erode and redeposit vast quantities of sediments, both offshore and onshore, heavily impacting both communities and ecosystems.



Tornadoes

- **Tornadoes** are mobile funnel-shaped rotating vortexes of wind that form and advance beneath large storm systems.
- Tornadoes are rapidly spinning columns of air that touch the ground from a thunderstorm, while cyclones are massive, swirling storm systems that form over warm ocean waters.
- Tornadoes vary considerably in their destructive power—how strong their winds are, how long they are in contact with the land surface, and the distance they travel.
- Storm systems can often produce multiple tornadoes (called a tornado outbreak). A single storm cell can sometimes produce multiple tornadoes simultaneously.

